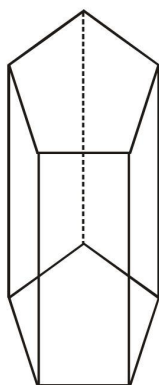


Mensuration – II

In the previous chapter we have dealt with properties of plane (two-dimensional) figures. Here, we will look at the areas and volumes of solids. Solids are three-dimensional object which in addition to areas, have volumes also.

In case of solids, we have two different types of areas.

- (i) **Lateral surface area or curved surface area:** Lateral surface area is the area of all surface of a solid excluding the top and the bottom surface. In solid (like cylinder, cone, sphere) where the lateral surface is curved, the lateral surface area is usually referred as the "curved surface area".
 - (ii) **Total surface area:** Total surface area is the area of all surfaces of a solid including the top and the bottom surface. So,
 Total surface area = Lateral surface area + Area of the top face + Area of the bottom face.
 There is a definite relationship between the number of vertices, the number of faces and the number of edges of any solid having regular polygonal faces.
 Let us look at the formula for surface areas and volumes of various solids.
- (1) **A right prism** is a solid whose top and bottom faces are parallel to each other and are identical polygons (of any number of sides).



The faces joining the top and bottom faces are rectangles and are called lateral faces.

There are as many lateral faces as there are sides in the base. The distance between the base and the top is called height or length of the right prism.

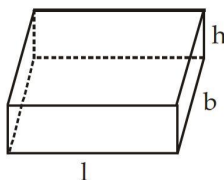
If a perpendicular is drawn from the centre of the top face then it passes through the centre of the base.

For any prism, Lateral surface area = perimeter of base $\times$ height of the prism.

Total surface area = Lateral surface area + $2 \times$ Area of base

Volume = Area of base $\times$ height of the prism

2. **Cuboid:** A cuboid is a right prism whose base is rectangle.



As shown in the figure, l and b are the length and breadth of the base of the cuboid and h is its height.

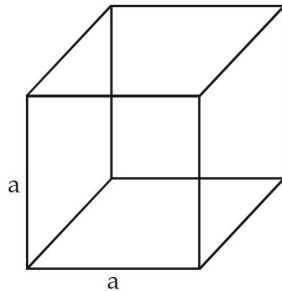
Volume = lbh

Lateral surface area = $2(l + b)h$

Total surface area = $2(lb + bh + lh)$

The length of the diagonal of the cuboid = $\sqrt{l^2 + b^2 + h^2}$

3. **Cube:**



A cube is a right prism whose base is square and height is equal to the side of the base.

Lateral surface area = $4a^2$ (Length of each edge = a)

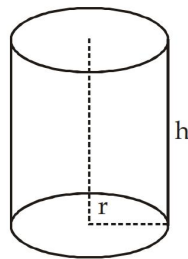
Total surface area = $6a^2$

Volume = a^3

The length of the longest diagonal of the cube = $a\sqrt{3}$. The longest diagonal of a cube is simply referred to as the diagonal of the cube.

Face diagonal of the cube = $\sqrt{2} a$ units.

4. **Right circular cylinder:**



r = radius of base
 h = height

A right prism whose base is a circle is a right circular cylinder. A cylinder has a single curved surface as its lateral faces.

Curved surface area = $2\pi rh$

Total surface area = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

Volume = $\pi r^2 h$

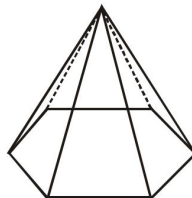
4. If R is the outer radius, r is the inner radius and h , is the height of a hollow cylinder, then

Volume of the material contained in it = $\pi(R^2 - r^2)h$

Curved surface area = $2\pi (R + r) h$

Total surface area = $2\pi (R + r) [h + (R - r)]$

5. **Pyramid:** A pyramid is a solid whose base is a polygon and whose faces are triangles. The common point where the triangle faces meet is called vertex. The perpendicular from the vertex to the base is called the height of the pyramid.



A right pyramid is a pyramid whose base is a regular polygon and the foot of perpendicular from the vertex to the base coincides with the centre of the base.

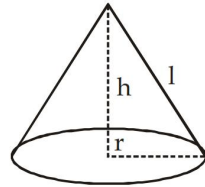
The length of the perpendicular from the vertex to any side of the base(along the slant lateral surface) is called the slant height of the prism.

Lateral surface area = $\frac{1}{2} \times \text{perimeter of the base} \times \text{slant height}$.

Total surface area = Lateral surface area + Area of the base.

Volume = $\frac{1}{3} \times \text{Area of base} \times \text{height}$

6. **Cone:**



r = radius of the base

h = height

l = slant height

$$l^2 = r^2 + h^2$$

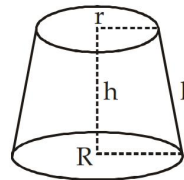
A cone is a right pyramid whose base is a circle. The lateral surface of a cone consists of a single curved surface.

Curved surface area = $\pi \cdot r \cdot l$

Total surface area = $\pi \cdot r \cdot l + \pi r^2 = \pi r(r + l)$

Volume = $\frac{1}{3} \pi r^2 h$

7. **Frustum of a cone:**



r = radius of top surface

R = radius of base

h = height

l = slant height

If a cone is cut into two parts by a plane parallel to the base, the portion containing the base is called the frustum of a cone.

Lateral surface area = $\pi(R + r)l$

Total surface area = $\pi(R^2 + r^2 + Rl + rl)$.

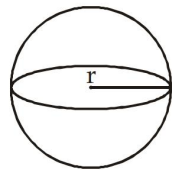
Volume = $\frac{1}{3} \pi(R^2 + Rr + r^2)h$

$$l^2 = (R - r)^2 + h^2$$

If H is the height of the cone from which the frustum is cut, then we have the following relationship:

$$\frac{r}{R} = \frac{H - h}{H}$$

8. A sphere is a solid figure formed by revolving a semi-circle on its diameter.



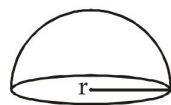
r = radius

Any point on the surface of a sphere is equidistant from the centre of the sphere and this distance is the radius of the sphere.

Surface area of a sphere = $4\pi r^2$

Volume of a sphere = $\frac{4}{3} \pi r^3$

9. **Hemisphere:**



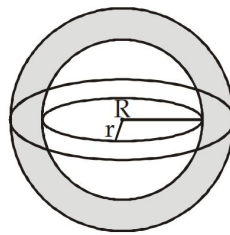
r = radius

Total surface area = $3\pi r^2$

Volume = $\frac{2}{3} \pi r^3$.

Curved surface area = $2\pi r^2$

10. **Spherical shell:**

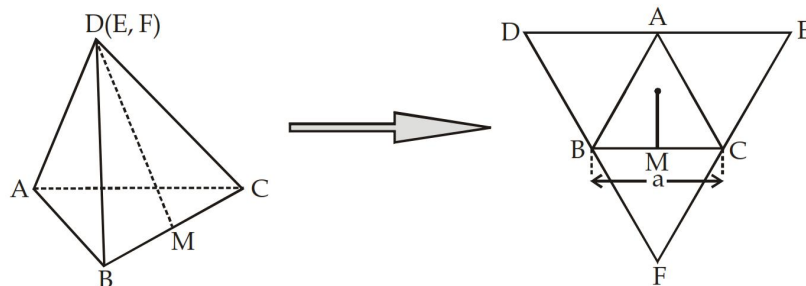


R = outer radius
r = inner radius

$$\text{Total surface area} = 4\pi(R^2 + r^2)$$

$$\text{Volume} = \frac{4}{3}\pi(R^3 - r^3)$$

11. **Tetrahedron:**



A tetrahedron is a pyramid with triangular base. It's a solid object with four triangular faces, three on the sides (lateral faces), and one on the bottom (base), and four vertices (corners). If the faces are all congruent equilateral triangle, then the tetrahedron is called regular.

In the above figure, we have a regular tetrahedron we obtain four equilateral triangles as shown.

$$\text{in-radius of } \triangle ABC = \frac{a}{2\sqrt{3}}$$

$$\text{Slant-height} = MF = \frac{\sqrt{3}}{2}a$$

$$\text{Height (h)} = \sqrt{\left(\frac{\sqrt{3}}{2}a\right)^2 - \left(\frac{a}{2\sqrt{3}}\right)^2} = \frac{\sqrt{2}}{\sqrt{3}}a.$$

$$(i) \text{ Volume (v)} = \frac{1}{3} \times (\text{Area of base}) \times \text{height}$$

$$= \frac{1}{3} \times \frac{\sqrt{3}}{4}a^2 \times \frac{\sqrt{2}}{\sqrt{3}}a = \frac{\sqrt{2}}{12}a^3.$$

$$(ii) \text{ Lateral surface area} = \frac{1}{2} \times \text{perimeter of the base} \times \text{slant height}.$$

$$= \frac{1}{2} \times 3a \times \frac{\sqrt{3}}{2}a = \frac{3\sqrt{3}}{4}a^2.$$

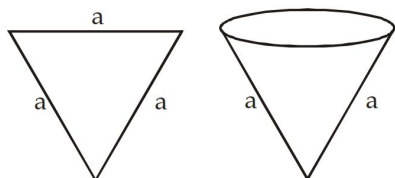
$$(iii) \text{ Total surface area} = \text{Lateral surface area} + \text{base area}$$

$$= \frac{3\sqrt{3}}{4}a^2 + \frac{\sqrt{3}}{4}a^2 = \sqrt{3}a^2.$$

Types of Questions

1. Each edge of an equilateral triangle is 'a' cm. A cone is formed by joining any two sides of the triangle. What is the radius and slant height of the cone?

Sol.



Slant height = a

$$2\pi r = a, \therefore \text{radius } (r) = \frac{a}{2\pi}$$

2. A solid metal ball of diameter 8 cm is melted and cast into smaller balls, each of radius 1 cm. The number of such balls is :

Sol. Since volume is constant

$$\therefore n \times \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi(4)^3$$

$$\Rightarrow n = 64$$

3. A circus tent is cylindrical up to a height of 3 m and conical above it. If its diameter is 105 m and the slant height of the conical part is 63 m, then the total area

of the canvas required to make the tent is $\left(\text{use } \pi \frac{22}{7}\right)$:

Sol. Total area of the canvas = $2\pi rh + \pi rl = \pi r(2h + l)$

$$= \frac{22}{7} \times \frac{105}{2} (2 \times 3 + 63)$$

$$= \frac{22}{7} \times \frac{105}{2} \times 69 = 11385 \text{ sq. metre}$$

4. The radii of the base of cylinder and a cone are in the ratio $\sqrt{3} : \sqrt{2}$ and heights in the ratio $\sqrt{2} : \sqrt{3}$. Their volumes are in the ratio of:

Sol. $\frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{\pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2}$

$$= 3 \cdot \left(\frac{r_1}{r_2}\right)^2 \cdot \left(\frac{h_1}{h_2}\right) = 3 \times \left(\frac{\sqrt{3}}{\sqrt{2}}\right)^2 \times \frac{\sqrt{2}}{\sqrt{3}}$$

$$= 3 \times \frac{\sqrt{3}}{\sqrt{2}} = 3\sqrt{3} : \sqrt{2}$$

5. A sphere of 20 cm radius is dropped into a cylindrical vessel of 60 cm diameter, which is partly filled with water, then its level rises by x cm. Find x:

Sol. Volume of water displaced = volume of sphere

$$\pi(30)^2 \times x = \frac{4}{3}\pi(20)^3$$

$$\Rightarrow x = \frac{320}{27} = 11\frac{23}{27} \text{ cm}$$

6. 125 identical cubes are cut from a big cube of edge 5cm and all the smaller cubes are arranged in a row to form a long cuboid. What is the percentage increase in the total surface area of the cuboid over total surface area of the cube?

Sol. Area of large cube = $6(5)^2$

$$\text{Area of cuboid} = 2(1 \times 1 + 1 \times 125 + 125 \times 1)$$

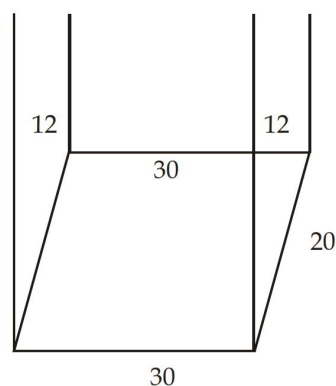
$$= 502 \text{ sq. units}$$

$\therefore$ % increase in surface area

$$= \frac{502 - 150}{150} \times 100 = 234\frac{2}{3} \%$$

7. A water tank is 30 cm long, 20 cm wide and 12m deep. It is made of iron sheet which is 3 m wide. The tank is open at the top. If the cost of iron sheet is Rs. 10per meter. Find the total cost of iron required to build the tank?

Sol.



Total surface area of tank

$$\text{TS A} = 30 \times 20 + 2(12 \times 20) + 2(30 \times 12) = 1800$$

$$\therefore \text{area of iron sheet} = 1800$$

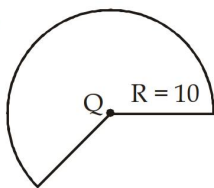
$$\Rightarrow \text{Length} \times \text{width} = 1800$$

$$\Rightarrow \text{Length} = \frac{1800}{3} = 600 \text{ m}$$

$$\Rightarrow \text{Cost} = 600 \times 10 = \text{Rs. } 6000.$$

8. From a circular sheet of paper, radius 10 cm, A sector of area 40% of the sheet is removed. If the remaining part is used to make a conical surface. Then the ratio of radius and height will be:

Sol.



Let radius of conical surface = r

$$\therefore 2\pi r = 40\% \text{ of } 2\pi R$$

$$\Rightarrow r = \frac{2}{5} \times 10 = 4 \text{ cm}$$

and slant height of cone = $l = R = 10 \text{ cm}$

$$\therefore \text{height } (h) = \sqrt{l^2 - r^2} = \sqrt{(10)^2 - (4)^2} = \sqrt{84} \text{ cm}$$

$$\therefore r : h = 4 : \sqrt{84}$$

9. An iron pipe 20 cm long has exterior diameter 25 cm. If the thickness of the pipe is 1 cm, then the whole surface area of the pipe?

Sol. Total surface area of pipe (hollow cylinder)

$$= 2\pi(R + r)[h + (R - r)]$$

here, $R - r$ = thickness = 1 cm, $h = 20 \text{ cm}$

$$R = \frac{25}{2} = 12.5 \text{ cm}$$

$$\therefore r = 12.5 - 1 = 11.5$$

$$\begin{aligned} \text{Area} &= 2 \times \frac{22}{7} (12.5 + 11.5)(20 + 1) \\ &= 44 \times 72 = 3168 \text{ cm}^2 \end{aligned}$$

10. A right circular cylinder and a cone have equal base radius and equal heights. If their curved surfaces are in the ratio 8 : 5, then the radius of the base and height are in the ratio:

Sol. Radius of the base = r units and height = h units

$$\Rightarrow \frac{\text{Curved surface of cylinder}}{\text{Curved surface of cone}} = \frac{2\pi rh}{\pi rl}$$

$$\Rightarrow \frac{8}{5} = \frac{2h}{l} \Rightarrow \frac{4}{5} = \frac{h}{\sqrt{h^2 - r^2}}$$

$$\Rightarrow \frac{16}{25} = \frac{h^2}{h^2 + r^2} \Rightarrow \frac{h^2 + r^2}{h^2} = \frac{25}{16}$$

$$\Rightarrow 1 + \frac{r^2}{h^2} = \frac{25}{16} \Rightarrow \frac{r^2}{h^2} = \frac{25}{16} - 1 = \frac{9}{16}$$

$$\Rightarrow \frac{r}{h} = \frac{3}{4}$$

Foundation

Questions

- If the length, breadth and height of a cuboid are 2 m, 2 m and 1 m respectively, then its surface area (in m^2) is:
(a) 8 (b) 12
(c) 16 (d) 24
- A rectangular sand box is 5 m wide and 2 m long. How many cubic metres of sand are needed to fill the box upto a depth of 10 cm?
(a) 1 (b) 10
(c) 100 (d) 1000
- A beam 9 m long, 40 cm wide and 20 cm deep is made up of iron which weight 50 kg per cubic metre. The weight of the beam is:
(a) 27 kg (b) 36 kg
(c) 48 kg (d) 56 kg
- The length of the longest rod that can be placed in a room 30 m long, 24 m broad and 18 m high, is:
(a) 30 m (b) $15\sqrt{2} \text{ m}$
(c) 60 m (d) $30\sqrt{2} \text{ m}$
- The perimeter of one face of a cube is 40 cm. The volume of the cube (in cm^3) is:
(a) 1600 (b) 1000
(c) 800 (d) 160
- If 1 cm^3 of silver weighs 10 gms and 8 gms of silver costs Rs. 30, then the cost of a silver cube of edge 4cm, is:
(a) Rs. 800 (b) Rs. 1200
(c) Rs. 1920 (d) Rs. 2400
- The volume of a cube with surface area 384 sq. cm, is:
(a) 215 cm^3 (b) 256 cm^3
(c) 484 cm^3 (d) 512 cm^3
- If the length of diagonal of a cube is $4\sqrt{3} \text{ cm}$, then the length of its edge is:
(a) 2 cm (b) 3 cm
(c) 4 cm (d) 6 cm
- If the areas of 3 adjacent faces of a cuboid are x, y, z respectively, then the volume of the cuboid is:
(a) xyz (b) $2xyz$
(c) $\sqrt{xyz}$ (d) $3\sqrt{xyz}$
- Two cubes each with 8 cm edge are joined end to end. The surface area of the resulting cuboid is:
(a) 1440 cm^2 (b) 830 cm^2
(c) 640 cm^2 (d) 560 cm^2

11. If three cubes of metal whose edges are 6 cm, 8 cm and 10 cm are melted and made into a single cube, then the edge of the new cube so formed will be:
 - (a) 6 cm
 - (b) 8 cm
 - (c) 12 cm
 - (d) 24 cm
12. A cube of side 6 cm is cut to a number of cubes each of side 2 cm. The number of cubes will be:
 - (a) 6
 - (b) 9
 - (c) 12
 - (d) 27
13. The volume of the cylinder whose height is 14 cm and diameter of base 4 cm, is:
 - (a) 176 cm^3
 - (b) 196 cm^3
 - (c) 276 cm^3
 - (d) 352 cm^3
14. If the diameter of the base of a cylindrical pillar is 4 m and its height is 21 m, then the cost of construction of the pillar at Rs. 1.50 per cubic metre is:
 - (a) Rs. 396
 - (b) Rs. 400
 - (c) Rs. 410
 - (d) Rs. 420
15. A right circular cylindrical tunnel of diameter 2 m and length 40 m is to be constructed from a sheet of iron. The area of the sheet required in m^2 , is:
 - (a) 40π
 - (b) 80π
 - (c) 160π
 - (d) 200π
16. If the curved surface area of a cylinder is 1760 sq. cm and its base radius is 14 cm, then its height is:
 - (a) 10 cm
 - (b) 15 cm
 - (c) 20 cm
 - (d) 40 cm
17. The volume (in cm^3) of a right circular cone of height 12 cm and base radius 6 cm, is:
 - (a) 12π
 - (b) 36π
 - (c) 72π
 - (d) 144π
18. If the radius of the base of a right circular cone is $3r$ and its height is equal to the radius of the base, then its volume is:
 - (a) $\frac{1}{3}\pi r^3$
 - (b) $\frac{2}{3}\pi r^3$
 - (c) $3\pi r^3$
 - (d) $9\pi r^3$
19. If the slant height and the radius of the base of a right circular cone are H and r respectively, then the ratio of the area of the lateral surface and the base is:
 - (a) $2H : r$
 - (b) $H : r$
 - (c) $H : 2r$
 - (d) $H^2 : r^2$
20. A conical tent has 60° angle at the vertex. The ratio of its radius and slant height is:
 - (a) 1 : 2
 - (b) 1 : 3
 - (c) $1 : \sqrt{3}$
 - (d) $1 : \sqrt{2}$
21. The ratio of the volume of a right circular cylinder and a right circular cone of the same base and height will be:
 - (a) 1 : 3
 - (b) 3 : 1
 - (c) 4 : 3
 - (d) 3 : 4
22. A solid metal ball of radius 8 cm is melted and cast into smaller balls, each of radius 2 cm. The number of such balls is:
 - (a) 8
 - (b) 16
 - (c) 32
 - (d) 64
23. The total surface area (in cm^2) of a solid hemisphere whose diameter is 14 cm, is:
 - (a) 588π
 - (b) 392π
 - (c) 147π
 - (d) 98π
24. The number of solid spheres, each of diameter 6 cm that could be moulded from a solid metal cylinder of height 45 cm and diameter 4 cm, is:
 - (a) 3
 - (b) 4
 - (c) 5
 - (d) 6
25. If a solid sphere of radius r is melted and cast into the shape of a solid cone of height r , then the radius of the base of the cone is:
 - (a) $2r$
 - (b) $3r$
 - (c) r
 - (d) $4r$
26. A tank 16 m long and 24 m wide contains water. How many cubic metres of water must be rushed into it to make the surface rise by $16\frac{2}{3} \text{ cm}$?
 - (a) 48 m^3
 - (b) 40 m^3
 - (c) 64 m^3
 - (d) 42 m^3
27. How many bricks each measuring 250 cm by 12.5 cm by 7.5 cm will be required to build a wall 5 m long, 3 m high and 20 cm thick?
 - (a) 148
 - (b) 128
 - (c) 168
 - (d) 158
28. How many bricks are required to build a wall 15 m long 3 m high and 50 cm thick, if each brick measures 25 cm by 12 cm by 6 cm?
 - (a) 16500
 - (b) 14500
 - (c) 12500
 - (d) 10500
29. How many small cubical blocks of side 5 cm can be cut from a cubical block whose each edge measures 20 cm?
 - (a) 56
 - (b) 48
 - (c) 64
 - (d) 52
30. Two cubes, each of side 12 cm, are joined end to end. The surface area of the resulting cuboid is:
 - (a) 1240 cm^2
 - (b) 1440 cm^2
 - (c) 2250 cm^2
 - (d) 4252 cm^2
31. A hemisphere of lead with radius 6 cm, is cast into a right circular cone of height 75 cm. The radius of the base of the cone is:
 - (a) 1.4 cm
 - (b) 2.4 cm
 - (c) 1.6 cm
 - (d) 3.2 cm

32. The radii of two cylinders are in the ratio 2 : 3 and their height are in the ratio 5 : 3. Ratio of their volume is:
 (a) 20 : 27 (b) 10 : 9
 (c) 18 : 13 (d) 9 : 20
33. A cylindrical iron rod is 70 cm long and the diameter of its end is 2 cm. What is its weight, reckoning a cubic centimeter of iron weight 10 grams?
 (a) 4 kg (b) 4.2 kg
 (c) 2.2 kg (d) Data inadequate
34. The perpendicular height of a conical tent is $\frac{14}{3}$ m and the diameter of its base is 6 metres. If 11 persons can sleep in this tent, find how many average m^3 of air each person gets?
 (a) 2 m^3 (b) 4 m^3
 (c) 6 m^3 (d) 8 m^3
35. The circumference of the base of a 9 m high conical tent is 44 m. The volume of the air contained in it is:
 (a) 462 m^3 (b) 452 m^3
 (c) 472 m^3 (d) 512 m^3
36. The curved surface of a circular cylinder of height 'h' and the slant surface of the cone of slant height '2h' having the same circular base are in the ratio of:
 (a) 1 : 1 (b) 1 : 2
 (c) 3 : 2 (d) 1 : 3
37. If the height of the cone is doubled then its volume is increased by:
 (a) 100% (b) 200%
 (c) 300% (d) 400%
38. Three cubes of side 3, 4 and 5, respectively are melted to form a new cube. The side of the new cube is:
 (a) 5 cm (b) 6 cm
 (c) 6.5 cm (d) 7 cm
39. The height and base radius of a cone are each increased by 100%. The volume of the cone now becomes:
 (a) double the original
 (b) four times the original
 (c) three times the original
 (d) eight times the original
40. The radius of a cylinder is increased by 20%, keeping its height unchanged. What is the percentage increase in its volume?
 (a) 33% (b) 44%
 (c) 22% (d) None of these

Moderate

1. Which of the following pairs is not correctly matched:

Geometrical objects	No. of Vertices
(a) Tetrahedron	4
(b) Pyramid with rectangular base	5
(c) Cube	6
(d) Triangle	3
2. A wooden box of dimensions $8\text{m} \times 7\text{m} \times 6\text{m}$ is to carry rectangular boxes of dimensions $8\text{cm} \times 7\text{cm} \times 6\text{cm}$. The maximum number of boxes that can be carried in the wooden box, is:
 (a) 9800000 (b) 7500000
 (c) 1000000 (d) 1200000
3. The volume of a cube (in cubic metres), whose diagonal is 2.5 m, is:
 (a) $\frac{125\sqrt{3}}{72}$ (b) $\frac{625}{8}$
 (c) $\frac{125\sqrt{2}}{32}$ (d) $\frac{125}{8}$
4. If the areas of the adjacent faces of a rectangular block are in the ratio 2 : 3 : 4 and its volume is 9000 cm^3 , then the length of the shortest side is:
 (a) 30 cm (b) 20 cm
 (c) 15 cm (d) 10 cm
5. The length, breadth and height of a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 100%, 200% and 200% respectively, the increase in the volume of the cuboid is:
 (a) 5 times (b) 6 times
 (c) 12 times (d) 17 times
6. If the diameter of the base of a closed right circular cylinder be equal to its height h, then its whole surface area is:
 (a) $2\pi h^2$ (b) $\frac{3}{2}\pi h^2$
 (c) $\frac{4}{3}\pi h^2$ (d) πh^2
7. Two steel sheets each of length a_1 and breadth a_2 are used to prepare the surface of two right circular cylinders—one having volume V_1 and height a_2 and the other having volume V_2 and height a_1 . Then:
 (a) $V_1 = V_2$ (b) $a_1 V_1 = a_2 V_2$
 (c) $a_2 V_1 = a_1 V_2$ (d) $\frac{V_1}{a_1} = \frac{V_2}{a_2}$
8. A rectangular tin sheets is 12 cm long and 5 cm broad. It is rolled along its length to form a cylinder by making the opposite edges just to touch each other. The volume of the cylinder (in cm^3) is:

- (a) $\frac{180}{\pi}$ (b) $\frac{120}{\pi}$
 (c) $\frac{100}{\pi}$ (d) $\frac{60}{\pi}$
9. The lateral surface of a cylinder is developed into a square whose diagonal is $\sqrt{5}$ cm. The area of the base of the cylinder (in sq. cm) is:
 (a) $\frac{5}{8\pi}$ (b) $\frac{5}{5\pi}$
 (c) $\frac{3}{8\pi}$ (d) $\frac{8}{3\pi}$
10. If the height of two cones are in the ratio of 1 : 4 and the ratio of their bases are in the ratio 4 : 1, then the ratio of their volumes is:
 (a) 1 : 2 (b) 2 : 3
 (c) 3 : 4 (d) 4 : 1
11. If the height and the radius of a cone are doubled, the volume of the cone becomes:
 (a) 3 times (b) 4 times
 (c) 6 times (d) 8 times
12. A conical tent has 60° at the vertex. If the curved surface is 100 m^2 , then the volume of the tent is:
 (a) $\frac{500}{\sqrt{6\pi}} \text{ m}^3$ (b) $\frac{500}{\sqrt{6}} \text{ m}^3$
 (c) $\frac{500}{\sqrt{\pi}} \text{ m}^3$ (d) None of these
13. From a cylindrical piece of wood of radius 4 cms and height 5 cms, a right circular cone with same base radius and height 3 cms is carved out. The surface area of the remaining wood is:
 (a) 56π (b) 60π
 (c) 76π (d) 92π
14. If a cone is cut into two parts by a horizontal plane passing through the mid-point of its axis, the ratio of the volumes of the upper part and the cone is:
 (a) 1 : 2 (b) 1 : 4
 (c) 1 : 6 (d) 1 : 8
15. The volume of a spherical shell whose internal and external diameters are 8 cm and 10 cm respectively (in cubic cm) is:
 (a) $\frac{122\pi}{3}$ (b) $\frac{244\pi}{3}$
 (c) 212 (d) 257
16. If the surface area of a sphere is $144\pi \text{ sq. m}$, then its volume (in cu. m) is:
 (a) 288π (b) 316π
 (c) 300π (d) 188π
17. If the height and diameter of a right circular cylinder are 32 cm and 6 cm respectively, then the radius of the sphere whose volume is equal to the volume of the cylinder is:
 (a) 3 cm (b) 4 cm
 (c) 6 cm (d) None of these
18. A right circular cylinder and a sphere have the same volume and same radius. The ratio of the area of their curved surfaces is:
 (a) 1 : 1 (b) 1 : 2
 (c) 1 : 3 (d) 2 : 3
19. A cone, a hemi-sphere and a cylinder stand on equal bases of radius R and have equal heights H. Their whole surfaces are in the ratio:
 (a) $(\sqrt{2} + 1) : 3 : 4$ (b) $(\sqrt{3} + 1) : 3 : 4$
 (c) $(\sqrt{2} + 1) : 7 : 8$ (d) $(\sqrt{3} + 1) : 7 : 8$
20. The ratio between the volume of a sphere and the volume of the circumscribing right cylinder is:
 (a) 2 : 1 (b) 1 : 1
 (c) 2 : 3 (d) 1 : 2
21. If the height of a cone is half the radius of a sphere, then the radius of the base of the cone which has the same volume as a sphere of radius 5 cm, is:
 (a) 10 cm (b) $10\sqrt{2} \text{ cm}$
 (c) $\frac{5\sqrt{2}}{2} \text{ cm}$ (d) $5\sqrt{2} \text{ cm}$
22. If a cylindrical rod of iron whose length is 12 times its radius is melted and cast into spherical balls of the same radius, then the number of balls will be:
 (a) 3 (b) 6
 (c) 9 (d) 27
23. A solid is hemi-spherical at the bottom and conical above. If the surface areas of the two parts are equal, then the ratio of its radius and the height of its conical part is:
 (a) 1 : 3 (b) $1 : \sqrt{3}$
 (c) 1 : 1 (d) $\sqrt{3} : 1$
24. A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end and the other end as its base. The volumes of the cylinder, hemisphere and the cone are respectively in the ratio of:
 (a) 2 : 3 : 2 (b) 3 : 2 : 1
 (c) 3 : 1 : 2 (d) 1 : 2 : 3
25. If all the edges of a rectangular prism are doubled and its shape is left unchanged, then its volume becomes:
 (a) 2 times (b) 3 times
 (c) 4 times (d) 8 times

26. If 210 m^3 of sand be thrown into a tank 12 m long and 5 m wide, find how much the water will rise?
 (a) 3.5 m (b) 4 m
 (c) 7 m (d) Data inadequate
27. How many cubes each of surface 24 cm^2 can be made out of a cube of edge measure one metre?
 (a) 165000 (b) 125000
 (c) 180000 (d) 155000
28. How many small cubes, each of 96 cm^2 surface area, can be formed from the material obtained by melting a larger cube of 384 cm^2 surface area?
 (a) 8 (b) 5
 (c) 800 (d) 8000
29. The circumference of one end of a frustum, of a right circular cone is 48 cm and of the other end is 34 cm. If the height of the frustum is 10 cm, its volume (in cubic centimeter) is:
 (a) 5400 (b) 1350
 (c) 2700 (d) 4050
30. How many coins 2 mm thick and 1.5 cm in diameter should be melted in order to form a right circular cylinder having base diameter 6 cm and height 8 cm?
 (a) 640 (b) 540
 (c) 740 (d) 840
31. A solid cylinder has a total surface area of 231 cm^2 .
 Its curved surface area is $\frac{2}{3}$ rd of the total surface area. Find the volume of the cylinder?
 (a) 270 cm^3 (b) 269.5 cm^3
 (c) 256.5 cm^3 (d) 289.5 cm^3
32. The radius of the base of a solid cylinder is 4 cm and its height is 3 cm. It is recasted into a cone of same radius, the height of the cone will be:
 (a) 3 cm (b) 6 cm
 (c) 9 cm (d) 27 cm
33. Find the length of the canvas 2 m in width required to make a conical tent 12 m in diameter and 6.3 m in slant height?
 (a) 118.8 m (b) 62.4 m
 (c) 59.4 m (d) 112.4 m
34. If base radius of a cone is increased by 20% and its slant height is made double, then by how much percent will the area of its curved surface be increased?
 (a) 140% (b) 160%
 (c) 130% (d) 180%
35. A cone of height 7 cm and base radius 3 cm is carved from a rectangular block of wood $10 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$. The percentage wood wasted is?
 (a) 34% (b) 46%
 (c) 54% (d) 66%
36. A rectangular sheet of area 264 cm^2 and width 11 cm is rolled along its breadth to make a hollow cylinder. The volume of the cylinder is?
 (a) 231 cm^3 (b) 230 cm^3
 (c) 235 cm^3 (d) 234 cm^3
37. The diameter of a sphere is 6 cm. It is melted and drawn into a wire of diameter 0.2 cm. Find the length of the wire?
 (a) 24 m (b) 28 m
 (c) 36 m (d) 32 m
38. The height of a cylinder is increased by 8%, keeping its radius unchanged. What is the percentage change in its volume?
 (a) 8% increase (b) 12% decrease
 (c) 8% decrease (d) None of these
39. Solid cone is 25 cm high and radius of its base is 50 cm. It is melted and re-cast into a solid sphere. Determine the surface area of the sphere?
 (a) 8757.28 cm^2 (b) 5877.42 cm^2
 (c) 7857.14 cm^2 (d) None of these
40. A rectangular tank is 30 m long 20 m broad. Water is being flown into it through a square pipe of side 5 cm. What is the speed of water if the level of water in the tank rises by 1 m in 8 hours?
 (a) 30 km/h (b) 36 km/h
 (c) 40 km/h (d) None of these

Difficult

1. Three solid iron cubes of edges 4 cm, 5 cm and 6 cm are melted together to make a new cube. 62 cm^3 of the melted material is lost due to improper handling. The area (in cm^2) of the whole surface of the newly formed cube is
 (a) 294 (b) 343
 (c) 125 (d) 216
2. The total surface area of a metallic hemisphere is 1848 cm^2 . The hemisphere is melted to form a solid right circular cone. If the radius of the base of the cone is the same as the radius of the hemisphere, its height is
 (a) 42 cm (b) 26 cm
 (c) 28 cm (d) 30 cm
3. A child reshapes a cone made up of clay of height 24 cm and radius 6 cm into a sphere. The radius (in cm) of the sphere is
 (a) 6 (b) 12
 (c) 24 (d) 48

4. Spherical Marbles of diameter 1.4 cm are dropped into a cylindrical beaker containing some water and are fully submerged. The diameter of the beaker is 7 cm. Find how many marbles have been dropped in it if the water rises by 5.6 cm?
- (a) 50 (b) 150
(c) 250 (d) 350
5. A right circular cone is exactly fitted inside a cube in such a way that the edges of the base of the cone are touching the edges of one of the faces of the cube and the vertex is on the opposite face of the cube. If the volume of the cube is 343 cc, what approximately is the volume of the cone?
- (a) 90 cc (b) 80 cc
(c) 70 cc (d) 60 cc
6. If h , c , v are respectively the height, curved surface area and volume of a right circular cone then the value of $3\pi v h^3 - c^2 h^2 + 9v^2$ is
- (a) 2 (b) -1
(c) 1 (d) 0
7. A ball of lead 4 cm in diameter is covered with gold. If the volume of the gold and lead are equal then find the thickness of the gold, [given $\sqrt[3]{2} = 1.259$] approximately?
- (a) 5.038 cm (b) 5.190 cm
(c) 1.038 cm (d) 0.518 cm
8. Sum of the length, width and depth of a cuboid is s and its diagonal is d . Its surface area is:
- (a) s^2 (b) d^2
(c) $s^2 - d^2$ (d) $s^2 + d^2$
9. Spherical ball of radius 1 cm is dropped into a conical vessel of radius 3 cm and slant height 6 cm. The volume of water (in cm^3), that can just immerse the ball, is
- (a) $\frac{5\pi}{3}$ (b) 3π
(c) $\frac{\pi}{3}$ (d) $\frac{4\pi}{3}$
10. Water flows at the rate of 10 metres per minute from cylindrical pipe 5 mm in diameter. How long it will take to fill up a conical vessel whose diameter at the base is 30 cm and depth 24 cm?
- (a) 28 minutes 48 seconds
(b) 51 minutes 12 seconds
(c) 51 minutes 24 seconds
(d) 28 minutes 36 seconds
11. The radius and the height of a cone are in the ratio 4:3. The ratio of the curved surface area to total surface area of the cone is:
- (a) 5 : 9 (b) 3 : 7
(c) 5 : 4 (d) 16 : 9
12. A right cylinder and a right circular cone have same radius and same volume. The ratio of height of the cylinder of that of the cone is:
- (a) 3 : 5 (b) 2 : 5
(c) 3 : 1 (d) 1 : 3
13. A large solid sphere is melted and moulded to form identical right circular cones with base radius and height same as the radius of the sphere. One of these cones is melted and moulded to form a smaller solid sphere. Then the ratio of the surface area of the sphere to the surface area of the larger sphere is
- (a) $1 : 3^{\frac{4}{3}}$ (b) $1 : 2^{\frac{3}{2}}$
(c) $1 : 3^{\frac{2}{3}}$ (d) $1 : 2^{\frac{4}{3}}$
14. A plane divides a right circular cone into two parts of equal volume. Then the ratio, in which the height of the cone is divided, is
- (a) $1 : \sqrt{2}$ (b) $1 : \sqrt[3]{2}$
(c) $1 : \sqrt[3]{2} - 1$ (d) $1 : \sqrt[3]{2} + 1$
15. A hemispherical cup of radius 4 cm is filled to the brim with coffee. The coffee is then poured into a vertical cone of radius 8 cm and height 16 cm. the percentage of the volume of the cone that remains empty is :
- (a) 87.5% (b) 80.5%
(c) 81.6% (d) 88.2%
16. There is a pyramid on a base which is a regular hexagon of side $2a$ cm. If every slant edge of this pyramid is of length $\frac{5a}{2}$ cm, then the volume of this pyramid is
- (a) $3a^2 \text{ cm}^3$ (b) $3\sqrt{2} a^2 \text{ cm}^3$
(c) $3\sqrt{3} a^3 \text{ cm}^3$ (d) $6a^3 \text{ cm}^3$
17. The base of a right pyramid is a square of side 16 cm long. If its height be 15 cm, then the area of the lateral surface of square in cm^2 is:
- (a) 136 (b) 544
(c) 800 (d) 1280
18. The base of a right pyramid is equilateral triangle of side $10\sqrt{3}$ cm. If the total surface area of the pyramid is $270\sqrt{3}$ sq. cm. Its height is:
- (a) $12\sqrt{3}$ cm (b) 10 cm
(c) $10\sqrt{3}$ cm (d) 12 cm

19. Base of a prism of height 10 cm is square. Total surface of the prism is 192 sq. cm. The volume of the prism is
 (a) 120 cm^3 (b) 640 cm^3
 (c) 90 cm^3 (d) 160 cm^3
20. Base of a right prism is a rectangle, the ratio of whose length and breadth is 3 : 2. If the height of the prism is 12 cm and total surface area is 288 sq. cm, the volume of the prism is:
 (a) 288 cm^3 (b) 290 cm^3
 (c) 286 cm^3 (d) 291 cm^3
21. An iron pillar has some part in the form of a right circular cylinder and remaining in the form of right circular cone. The radius of the base of the cone as well as cylinder is 21 cm. The cylindrical part is 80 cm high and conical part is 16 cm high. Find the weight of the pillar, if 1 cm^3 of iron weight 8.45 g :
 (a) 999.39 kg (b) 111 kg
 (c) 1001 kg (d) 989 kg
22. A hollow sphere of outer diameter 24 cm is cut into two equal hemisphere. The total surface area of one of the hemisphere is $1436\frac{2}{7} \text{ cm}^2$. Each one of the hemisphere is filled with water. What is the volume of water that can be filled in each of the hemisphere?
 (a) $3358\frac{2}{3} \text{ cm}^3$ (b) $3528\frac{2}{3} \text{ cm}^3$
 (c) $2359\frac{2}{3} \text{ cm}^3$ (d) $9335\frac{2}{3} \text{ cm}^3$
23. A big cube of side 8 cm is formed by rearranging together 64 small but identical cubes each of side 2 cm. Further, if the corner cubes in the topmost layer of the big cube are removed, what is the change in total surface area of the big cube?
 (a) 16 cm^2 , decreases
 (b) 48 cm^2 , decreases
 (c) 32 cm^2 , decreases
 (d) remains the same as previously
24. The volume of a cylinder is 48.125 cm^3 , which is formed by rolling a rectangular paper sheet along the length of the paper. If a cuboidal box (without any lid i.e., open at the top) is made from the same sheet of paper by cutting out the square of side 0.5 cm from each of the four corners of the paper sheet, then what is the volume of this box?
 (a) 20 cm^3 (b) 38 cm^3
 (c) 19 cm^3 (d) None of these
25. A blacksmith has a rectangular sheet of iron. He has to make a cylindrical vessel with both circular ends are closed. When he minimises the wastage of the sheet of iron, then what is the ratio of the wastage to the utilised area of sheet?
 (a) $\frac{1}{11}$ (b) $\frac{2}{17}$
 (c) $\frac{3}{22}$ (d) None of these
26. Varun needs an open box of capacity 864 m^3 . Actually where he lives, the rates of paints are soaring high so he wants to minimize the surface area of the box keeping the capacity of the box same as required. What is the base area and height of such a box?
 (a) 36 m^2 , 24m (b) 216 m^2 , 4m
 (c) 144 m^2 , 6m (d) None of these
27. Find the length of the string bound on a cylindrical tank whose base diameter and height are $5\frac{1}{11} \text{ cm}$ and 48 cm respectively. The string makes exactly four complete turns round the cylinder, while its two ends touch the tank's top and bottom:
 (a) 75 cm (b) 70 cm
 (c) 60 cm (d) 80 cm
28. Find the radius of maximum size of sphere which can be inscribed or put in a cone whose base radius and height are 6 cm and 8 cm respectively?
 (a) 4 cm (b) 5 cm
 (c) 3 cm (d) None of these
29. The base radius and height of a cone is 5 cm and 25 cm respectively. If the cone is cut parallel to its base at a height of h from the base. If the volume of this frustum is 110 cm^3 . Find the the radius of smaller cone?
 (a) $(104)^{1/3} \text{ cm}$ (b) $(104)^{1/2} \text{ cm}$
 (c) 5 cm (d) None of these
30. A rectangle based pyramid, length and width of the base is 18 cm and 10 cm respectively. Find the total surface area, if its height is 12 cm:
 (a) 267 cm^2 (b) 564 cm^2
 (c) 516 cm^2 (d) None of these

Previous Year (Memory Based)

1. If each side of a cube is increased by 10%, then the volume of the cube will increase by:
 (a) 30% (b) 10%
 (c) 33.1% (d) 25%
2. A cube of edge 5 cm is cut into cubes each of edge of 1 cm. The ratio of the total surface area of one of the small cubes to that of the large cube is equal to:
 (a) 1 : 125 (b) 1 : 5
 (c) 1 : 625 (d) 1 : 25
3. A parallelopiped whose sides are in ratio 1 : 2 : 4 have the same volume as a cube. The ratio of their surface area is:
 (a) 7 : 5 (b) 4 : 3
 (c) 8 : 5 (d) 7 : 6
4. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of the four walls, the volume of the hall (in m³) is:
 (a) 720 (b) 900
 (c) 1200 (d) 1800
5. The length, breadth and height of a cuboid are in the ratio 1 : 2 : 3. If they are increased by 100%, 200% and 300%, respectively then, compare to the original volume and the increase in the volume of the cuboid will be:
 (a) 5 times (b) 18 times
 (c) 12 times (d) 23 times
6. A cistern of capacity 8000 L measures externally 3.3 m by 2.6 m by 1.1 m and its walls are 5 cm thick. The thickness of the bottom is:
 (a) 1 m (b) 1.1 m
 (c) 1 dm (d) 90 cm
7. A wooden box of dimension 8 m × 7 m × 6 m is to carry rectangular boxes of dimensions 8 cm × 7 cm × 6 cm. The maximum number of boxes that can be carried in 1 wooden box is:
 (a) 7500000 (b) 9800000
 (c) 1200000 (d) 1000000
8. A solid sphere of 6 cm diameter is melted and recast into 8 solid spheres of equal volume. The radius (in cm) of each small sphere is:
 (a) 1.5 (b) 3
 (c) 2 (d) 2.5
9. The volume of two spheres are in the ratio 8 : 27. The ratio of their surface areas is:
 (a) 4 : 9 (b) 2 : 3
 (c) 4 : 5 (d) 5 : 6
10. The radius of a right circular cone is 3 cm and its height is 4 cm. The total surface area of the cone is:
 (a) 48.4 sq. cm (b) 64.4 sq. cm
 (c) 96.4 sq. cm (d) 75.4 sq. cm
11. The volume of a right circular cone is 1232 cm³ and its vertical height is 24 cm. Its curved surface area is:
 (a) 154 cm² (b) 550 cm²
 (c) 604 cm² (d) 704 cm²
12. If a right circular cone is separated into solids of volumes V₁, V₂ and V₃ by two planes parallel to the base, which also trisect the altitude, then V₁ : V₂ : V₃ is:
 (a) 1 : 2 : 3 (b) 1 : 4 : 6
 (c) 1 : 6 : 9 (d) 1 : 7 : 19
13. The radius of the base and height of a right circular cone are in the ratio 5 : 12. If the volume of the cone is $314\frac{2}{7}$ cm³, the slant height (in cm) of the cone will be:
 (a) 12 (b) 13
 (c) 15 (d) 17
14. The base of a conical tent is 19.2 m in diameter and its height is 2.8 m. The area (in sq. m) of the canvas required to put up such a tent is nearly $\left(\text{take } \pi = \frac{22}{7}\right)$
 (a) 3017.10 (b) 3170
 (c) 301.71 (d) 30.17
15. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, then volume of the cone?
 (a) increases by 25% (b) increases by 50%
 (c) remains unaltered (d) decreases by 25%
16. A cone is cut at mid-point of its height by a frustum parallel to its base. The ratio between the two parts of cone would be:
 (a) 1 : 1 (b) 1 : 8
 (c) 1 : 4 (d) 1 : 7
17. A right circular cone is cut off at the middle of its height and parallel to base. Call smaller cone thus formed A and remaining part B. Then:
 (a) Vol A < Vol B (b) Vol A = Vol B
 (c) Vol A > Vol B (d) Vol A = $\frac{\text{Vol B}}{2}$
18. A rectangular piece of paper of dimensions 22 cm by 12 cm is rolled along its length to form a cylinder. The volume (in cm³) of the cylinder so formed is:
 $\left(\text{take } \pi = \frac{22}{7}\right)$

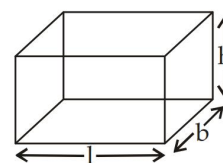
- (a) 562 (b) 412
(c) 462 (d) 362
19. The diameters of two cylinders, whose volumes are equal, are in the ratio 3 : 2. Their heights will be in the ratio:
(a) 4 : 9 (b) 5 : 6
(c) 5 : 8 (d) 8 : 9
20. The base radii of two cylinders are in the ratio 3 : 4 and their heights are in the ratio 8 : 7. The ratio of their volumes is:
(a) 27 : 20 (b) 9 : 14
(c) 9 : 4 (d) 4 : 9
21. A hollow cylindrical tube 20 cm long, is made of iron and its external and internal diameters are 8 cm and 6 cm, respectively. The volume of iron used in making the tube is: $\left(\text{use } \pi = \frac{22}{7} \right)$
(a) 1760 cm³ (b) 880 cm³
(c) 440 cm³ (d) 220 cm³
22. The height of the right pyramid whose area of the base is 30 m² and volume is 500 m³, is:
(a) 50 m (b) 60 m
(c) 40 m (d) 20 m
23. A prism has as the base a right angled triangle whose sides adjacent to the right angles are 10 cm and 12 cm long. The height of the prism is 20 cm. The density of the material of the prism is 6 gm/cm³. The weight of the prism is:
(a) 6.4 kg (b) 7.2 kg
(c) 3.4 kg (d) 4.8 kg
24. The base of a right pyramid is a square of side 16 cm long. If its height be 15 cm, then the area of the lateral surface (in cm²) is:
(a) 136 (b) 544
(c) 800 (d) 1280
25. The base of a solid right prism is a triangle whose sides are 9 cm, 12 cm and 15 cm. The height of the prism is 5 cm. Then, the total surface area of the prism is:
(a) 180 cm² (b) 234 cm²
(c) 288 cm² (d) 270 cm²
26. A sphere is placed inside a right circular cylinder so as to touch the top, base and the lateral surface of the cylinder. If the radius of the sphere is R, the volume of the cylinder is:
(a) 2πR³ (b) 4πR³
(c) 8πR³ (d) $\frac{8}{3}\pi R^2$
27. The base of a right prism is an equilateral triangle. If the lateral surface area and volume is 120 cm² and 40√3 cm³ respectively, then the side of base of the prism is:
(a) 4 cm (b) 5 cm
(c) 7 cm (d) 40 cm
28. A conical cup is filled with ice-cream. The ice-cream forms a hemispherical shape on its open top. The height of the hemispherical part is 7 cm. The radius of the hemispherical part equals the height of the cone. The, the volume of the ice-cream is: $\left[\pi = \frac{22}{7} \right]$
(a) 1078 cu cm (b) 1708 cu cm
(c) 7108 cu cm (d) 7180 cu cm
29. Two metallic right circular cones having their heights 3.8 cm and 4.6 cm and radii of their bases 2.1 cm each, have been melted together and recast into a sphere. The diameter (in cm) of the sphere is:
(a) 6.3 (b) 1.4
(c) 3.5 (d) 4.2
30. A solid wooden toy is in the shape of a right circular cone mounted on a hemisphere. If the radius of the hemisphere is 4.2 cm and the total height of the toy is 10.2 cm, find the volume of the wooden toy (nearly):
(a) 104 cm³ (b) 162 cm³
(c) 427 cm² (d) 266 cm³
31. The radii of the base of a cylinder and a cone are in the ratio √3 : √2 and their heights are in the ratio √2 : √3. Their volumes are in the ratio of:
(a) √3 : √2 (b) 3√3 : √2
(c) √3 : 2√2 (d) √2 : √6
32. A cylindrical can, whose base is horizontal and is of internal radius 3.5 cm, contains sufficient water, so that when a solid sphere is placed inside, water just covers the sphere. The sphere fits in the can exactly. The depth of water in a can before the sphere was put:
(a) $\frac{35}{3}$ cm (b) $\frac{17}{3}$ cm
(c) $\frac{7}{3}$ cm (d) $\frac{14}{3}$ cm
33. A solid consists of a circular cylinder with exact fitting right circular cone placed on the top. The height of the cone is h. If total volume of the solid is 3 times the volume of cone, then the height of circular cylinder is:
(a) 2h (b) 4h
(c) $\frac{3h}{2}$ (d) $\frac{2h}{3}$
34. From a right circular cylinder of radius 10 cm and height 21 cm, a right circular cone of same base-radius is removed. If the volume of the remaining portion is 4400 cm³, then the height of the removed cone $\left(\text{take } \pi = \frac{22}{7} \right)$ is:

- (a) 15 cm (b) 18 cm
(c) 21 cm (d) 24 cm
35. A cylindrical rod of iron whose height is eight times its radius is melted and cast into spherical balls each of half the radius of the cylinder. The number of such spherical balls is:
(a) 12 (b) 16
(c) 24 (d) 48
36. A sphere and a cube have equal surface areas. The ratio of the volume of the sphere to that of the cube is:
(a) $\sqrt{\pi} : \sqrt{6}$ (b) $\sqrt{6} : \sqrt{\pi}$
(c) $\sqrt{2} : \sqrt{\pi}$ (d) $\sqrt{\pi} : 3$
37. A copper wire of length 36 m and diameter 2 mm is melted to form a sphere. The radius of the sphere (in cm) is:
(a) 2.5 (b) 3
(c) 3.5 (d) 4
38. The ratio of the volume of a cube to that of a sphere, which will exactly fit inside the cube, is:
(a) $6 : \pi$ (b) $4 : \pi$
(c) $5 : 3\pi$ (d) $4 : 3$
39. A cone of height 15 cm and base diameter 30 cm is curved out of a wooden sphere of radius 15 cm. The percentage of wasted wood is:
(a) 75% (b) 50%
(c) 40% (d) 25%
40. Each of the measure of the radius of base of a cone and that of a sphere is 8 cm. Also, the volumes of these two solids are equal. The slant height of the cone is:
(a) $8\sqrt{17}$ cm (b) $4\sqrt{17}$ cm
(c) $34\sqrt{2}$ cm (d) 34 cm

Foundation

Solutions

1. (c); length = 2m, breadth = 2m and height = 1m
Total surface area = $2(lb + bh + hl)$
 $= 2(2 \times 2 + 2 \times 1 + 1 \times 2) = 2(4 + 2 + 2) = 16\text{m}^2$
2. (a); Volume = $l \times b \times h$
Volume = $5\text{m} \times 2\text{m} \times \frac{10}{100}\text{m} = \frac{100}{100}\text{m}^3 = 1\text{m}^3$
3. (b); Volume of beam = $900 \times 40 \times 20 = 720000\text{ cm}^3$
 $= .72\text{ m}^3$
Weight of the beam = $50 \times .72 = 36\text{ kg}$
4. (d); length of longest rod = $\sqrt{l^2 + b^2 + h^2}$
 $= \sqrt{30^2 + 24^2 + 18^2} = \sqrt{900 + 576 + 324}$
 $= \sqrt{900 + 900} = \sqrt{1800} = 30\sqrt{2}\text{ m}$
5. (b); Let the side of cube be a
Here, $4a = 40$, $a = 10\text{ cm}$
Volume of cube = $10^3 = 1000\text{ cm}^3$
6. (d); Volume of cube = $a^3 = 4^3 = 64\text{ cm}^3$
Weight of cube = $64 \times 10 = 640\text{ gms}$
Cost of 8 gms = Rs. 30
Cost of 640 gm = $\frac{30}{8} \times 640 = \text{Rs. } 2400$
7. (d); Surface area = $6a^2 = 384$
 $a^2 = \frac{384}{6} = 64$, $a = 8\text{ cm}$
Volume = $a^3 = 8^3 = 512\text{ cm}^3$
8. (c); Let the edge of the cube be 'a'.
Length of diagonal = $a\sqrt{3} = 4\sqrt{3}$, $a = 4\text{ cm}$
9. (c); Area of three adjacent faces are x, y and z respectively.
Let the side of cuboid be l, b and h.
therefore on multiplying
x, y and z
 $x \times y \times z = lb \times bh \times hl$
 $xyz = l^2b^2h^2$
 $= (lbh)^2$
 $lbh = \sqrt{xyz}$
Volume of cuboid = $lbh = \sqrt{xyz}$
10. (c); Surface area = $2(lb + bh + hl)$
 $= 2(16 \times 8 + 8 \times 8 + 8 \times 16)$
 $= 2(128 + 64 + 128)$
 $= 2 \times 320 = 640\text{ cm}^2$
11. (c); Volume of three cubes = $6^3 + 8^3 + 10^3$
 $= 216 + 512 + 1000$
 $= 1728\text{ cm}^3$
Volume of new cube = $a^3 = 1728$, $a = 12\text{ cm}$



12. (d); Number of cubes = $\frac{\text{Volume of big cube}}{\text{Volume of small cube}}$

$$= \frac{6 \times 6 \times 6}{2 \times 2 \times 2} = 27$$

13. (a); Height of cylinder = 14 cm

$$\text{radius of cylinder} = \frac{d}{2} = \frac{4}{2} = 2 \text{ cm}$$

$$\begin{aligned} \text{volume of cylinder} &= \pi r^2 h = \frac{22}{7} \times 2^2 \times 14 \\ &= 22 \times 8 = 176 \text{ cm}^3 \end{aligned}$$

14. (a); Volume of cylindrical pillar = $\pi r^2 h$

$$= \frac{22}{7} \times 2 \times 2 \times 21 = 264 \text{ m}^3$$

$$\text{Cost of construction} = \text{Rs. } 1.50 \times 264 = \text{Rs. } 396$$

15. (b); Radius of tunnel = $\frac{d}{2} = \frac{2}{2} = 1 \text{ m}$

$$\text{length} = 40 \text{ m}$$

$$\text{Area of sheet} = 2\pi rh = 2 \times \pi \times 1 \times 40 = 80\pi \text{ m}^2$$

16. (c); $2\pi rh = 1760$, $2\pi \times 14 \times h = 1760$

$$h = \frac{1760 \times 7}{2 \times 22 \times 14} = 20 \text{ cm}$$

17. (d); Volume = $\frac{1}{3} \pi r^2 h = \frac{1}{3} \times \pi \times 6 \times 6 \times 12 = 144\pi \text{ cm}^3$

18. (d); Volume of cone = $\frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (3r)^2 \times 3r = 9\pi r^3$

19. (b); Lateral surface area of cone = πrH

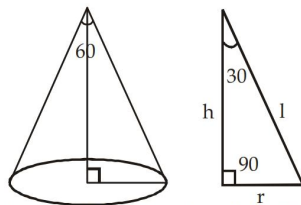
$$\text{Base area} = \pi r^2$$

$$\frac{\text{Lateral surface area}}{\text{Base area}} = \frac{\pi rH}{\pi r^2} = \frac{H}{r}$$

20. (a); $\frac{\text{Radius of base}}{\text{Slant height}} = \sin 30^\circ$

$$\frac{r}{l} = \sin 30^\circ$$

$$\frac{r}{l} = \frac{1}{2}$$



21. (b); Let radius of each be r and height of each be h.

$$\text{Then, } \frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{\pi r^2 h}{\frac{1}{3} \pi r^2 h} = \frac{3}{1}$$

22. (d); Number of balls = $\frac{\text{Volume of metal ball}}{\text{Volume of smaller ball}}$

$$= \frac{\frac{4}{3} \pi \times (8)^3}{\frac{4}{3} \pi \times (2)^3} = \frac{512}{8} = 64$$

23. (c); Total Surface area = $3\pi r^2$
 $= 3\pi \times 7^2 = 147\pi \text{ cm}^2$

24. (c); Volume of cylinder = $\pi r^2 h = \frac{22}{7} \times 2 \times 2 \times 45$

$$\text{Volume of sphere} = \frac{4}{3} \times \pi \times r^3 = \frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3$$

$$\text{Number of solid sphere} = \frac{\frac{22}{7} \times 2 \times 2 \times 45}{\frac{4}{3} \times \frac{22}{7} \times 3 \times 3 \times 3} = 5$$

25. (a); Let the radius of cone be R.

$$\text{Volume of sphere} = \text{Volume of cone}$$

$$\frac{4}{3} \pi r^3 = \frac{1}{3} \pi R^2 \cdot r, \quad R^2 = 4r^2, \quad R = 2r$$

26. (c); Volume of water = $16 \times 24 \times \frac{16 \frac{2}{3}}{100}$

$$= \frac{16 \times 24 \times 50}{3 \times 100} = 64 \text{ m}^3$$

27. (b); Number of bricks = $\frac{500 \times 300 \times 20}{250 \times 12.5 \times 7.5} = 128$

28. (c); Number of bricks = $\frac{1500 \times 300 \times 50}{25 \times 12 \times 6} = 12500$

29. (c); Edge of cubical block = 20 cm

$$\text{Volume} = 20^3 = 8000 \text{ cm}^3$$

$$\text{Edge of small cubical block} = 5 \text{ cm}$$

$$\text{Volume} = 5^3 = 125 \text{ cm}^3$$

$$\text{Number of smaller cubes} = \frac{8000}{125} = 64$$

30. (b); Surface area = $2(12 \times 24 + 12 \times 12 + 12 \times 24)$
 $= 2(288 + 144 + 288)$
 $= 2 \times 720 = 1440 \text{ cm}^2$

31. (b); Volume of hemisphere = Volume of cone

$$\frac{2}{3} \pi r^3 = \frac{1}{3} \pi R^2 H, \quad \frac{2}{3} \times 6 \times 6 \times 6 = \frac{1}{3} \times R^2 \times 75$$

$$R^2 = \frac{2 \times 6 \times 6 \times 6}{75} = 5.76, \quad R = \sqrt{5.76} = 2.4 \text{ cm}$$

32. (a); Let the radii of two cylinders be $2x$ and $3x$ and their height $= 5y$ and $3y$

$$\text{Required ratio} = \frac{\pi(2x)^2(5y)}{\pi(3x)^2(3y)} = \frac{20}{27}$$

33. (c); Volume of iron rod $= \pi r^2 h$

$$= \frac{22}{7} \times 1 \times 1 \times 70 = 220 \text{ cm}^3$$

$$\therefore \text{Weight of the cylinder} = \frac{220 \times 10}{1000} = 2.2 \text{ kg}$$

34. (b); Here, $h = \frac{14}{3} \text{ m}$, $r = 3 \text{ m}$

$$\text{Volume} = \frac{1}{3} \times \frac{22}{7} \times \frac{14}{3} \times 3 \times 3 = 44 \text{ m}^3$$

$$\text{Average of air each person get} = \frac{44}{11} = 4 \text{ m}^3$$

35. (a); Circumference $= 44 \text{ m}$

$$2\pi r = 44$$

$$r = \frac{44}{2 \times 22} \times 7 = 7 \text{ m}$$

$$\text{Volume} = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 9 = 462 \text{ cm}^3$$

36. (a); Curved surface area of cylinder $= 2\pi rh$
Curved surface area of cone $= \pi rl$

$$\text{Required ratio} = \frac{2\pi rh}{\pi r(2h)} = 1 : 1$$

37. (a); Original volume $= \frac{1}{3} \pi r^2 h$

$$\text{New volume} = \frac{1}{3} \pi r^2 (2h) = \frac{2}{3} \pi r^2 h$$

$$\text{Increase \%} = \frac{\frac{1}{3} \pi r^2 h - \frac{1}{3} \pi r^2 h}{\frac{1}{3} \pi r^2 h} \times 100 = 100\%$$

38. (b); Volume of cubes with sides 3, 4 and 5 cm are 3^3 , 4^3 and 5^3 respectively.

$$\text{Total volume} = 3^3 + 4^3 + 5^3 = 216 \text{ cm}^3$$

$$\text{Side of cube} = \sqrt[3]{216} = 6 \text{ cm}$$

39. (d); Let the radius of cone $= r$, height $= h$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$\text{Increased radius} = 2r, \text{ height} = 2h$$

$$\text{Increased volume} = \frac{1}{3} \pi (2r)^2 (2h) = \frac{8}{3} \pi r^2 h$$

$$= 8 \left(\frac{1}{3} \pi r^2 h \right)$$

$$= 8 \text{ times the original volume}$$

40. (b); Percentage increase $= \left[x + y + \frac{xy}{100} \right] \%$

$$= \left[20 + 20 + \frac{20 \times 20}{100} \right] \% = (40 + 4) \% = 44 \%$$

Moderate

1. (c); The vertices of a tetrahedron, pyramid, cube and a triangle are 4, 5, 8 and 3 respectively.
Therefore, (c) option is not correctly matched.

2. (c); Number boxes $= \frac{\text{Volume of wooden box in cm}^3}{\text{Volume of 1 brick in cm}^3}$

$$= \frac{800 \times 700 \times 600}{8 \times 7 \times 6} = 1000000$$

3. (a); Diagonal of cube $= 2.5 \text{ m}$

$$\text{Side cube} = \frac{2.5}{\sqrt{3}} \text{ m}$$

$$\text{Volume} = a^3 = \left(\frac{2.5}{\sqrt{3}} \right)^3 = \left(\frac{2.5 \times 2.5 \times 2.5}{\sqrt{3} \times \sqrt{3} \times \sqrt{3}} \right) \text{ m}^3$$

4. (c); Let

$$= \left(\frac{125}{24\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) \text{ m}^3 = \frac{125\sqrt{3}}{72} \text{ m}^3$$

$$lb = 2x, \quad bh = 3x \quad \text{and} \quad lb = 4x$$

$$24x^3 = l^2 b^2 h^2 = (lbh)^2 = 9000 \times 9000$$

$$x^3 = \frac{9000 \times 9000}{24} = 3375000$$

$$x = 150 \text{ cm}$$

$$lb = 300, \quad bh = 450, \quad lh = 600$$

$$h = 30, \quad l = 20, \quad b = 15$$

$$\text{Smallest side } b = 15 \text{ cm}$$

5. (d); Let original length $= x$

$$\text{breadth} = 2x \text{ and height} = 3x$$

$$\text{New length} = 200\% \text{ of } x = 2x$$

New breadth = 300% of $2x = 6x$

New height = 300% of $3x = 9x$

Original volume = $6x^3$, New volume = $108x^3$

Increase in volume = $\frac{102x^3}{6x^3}$ times = 17 times

6. (b); Here diameter, d = height, h

$$T.S.A = 2\pi rh + 2\pi r^2$$

$$= \pi(2r)h + \frac{\pi(2r)^2}{2} = \pi h^2 + \frac{\pi h^2}{2} = \frac{3\pi h^2}{2}$$

7. (c); Lateral surface area = $2\pi r_1 a_2 = a_1 a_2$, $r_1 = \frac{a_1}{2\pi}$

$$\text{Lateral Surface area} = 2\pi r_2 a_1 = a_1 a_2, \quad r_2 = \frac{a_2}{2\pi}$$

$$V_1 = \pi r_1^2 a_2 = \pi \times \frac{a_1^2}{4\pi^2} \times a_2 = \frac{a_1^2 a_2}{4\pi}$$

$$V_2 = \pi r_2^2 a_1 = \pi \times \frac{a_2^2}{4\pi^2} \times a_1 = \frac{a_2^2 a_1}{4\pi}$$

$$a_2 V_1 = a_1 V_2$$

8. (a); When sheet is rolled along its length, height = 5 cm

$$2\pi r = 12, \quad r = \frac{6}{\pi}$$

$$\text{Volume} = \pi r^2 h = \left(\pi \times \frac{36}{\pi^2} \times 5 \right) = \frac{180}{\pi} \text{ cm}^3$$

9. (a); Circumference of base = height

Lateral surface area of cylinder = Area of square

$$2\pi rh = \frac{1}{2} \times (\sqrt{5})^2 \Rightarrow (2\pi r) \times (2\pi r) = \frac{5}{2}$$

$$\pi^2 r^2 = \frac{5}{8} \Rightarrow \pi r^2 = \frac{5}{8\pi} \text{ cm}^2$$

10. (d); Let their height be x and $4x$ and let their radii be $4y$ and y .

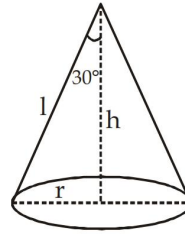
$$\text{Ratio of their volumes} = \frac{\frac{1}{3} \pi (4y)^2 \times x}{\frac{1}{3} \pi y^2 \times 4x} = \frac{4}{1}$$

11. (d); Let original radius = R and original height = H
New radius = $2R$ and New height = $2H$

$$\frac{\text{New volume}}{\text{Original volume}} = \frac{\frac{1}{3} \pi (2R)^2 \times 2H}{\frac{1}{3} \pi R^2 \times H} = \frac{8}{1}$$

So, it becomes 8 times.

12. (a);



$$\pi r l = 100, \quad \sin 30^\circ = \frac{r}{l} = \frac{1}{2}$$

$$l = 2r$$

$$\pi r \times 2r = 100$$

$$\pi r^2 = 50, \quad r = \frac{5\sqrt{2}}{\sqrt{\pi}}$$

$$\text{Also, } \frac{h}{r} = \cot 30^\circ = \sqrt{3}$$

$$h = \sqrt{3}r, \quad h = \frac{\sqrt{3} \times 5\sqrt{2}}{\sqrt{\pi}} = \frac{5\sqrt{6}}{\sqrt{\pi}}$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \cdot \frac{50 \times 5\sqrt{6}}{\sqrt{\pi}} = \frac{500}{\sqrt{6\pi}} \text{ m}^3$$

13. (c); Required area = curved surface of the cylinder + Area of top curved surface area of cone.

$$= 2\pi \times 4 \times 5 + \pi \times 4 \times 4 + \pi \times 4 \times 5$$

$$= 40\pi + 16\pi + 20\pi = 76\pi$$

14. (d); Let R be the radius and H be the height of the cone.

$$\text{Its volume} = \frac{1}{3} \pi R^2 H = V$$

$$\text{Radius of upper part} = \frac{R}{2} \text{ and Its height} = \frac{H}{2}$$

$$\text{Volume of upper part} = \frac{1}{3} \pi \times \left(\frac{R}{2} \right)^2 \times \left(\frac{H}{2} \right)$$

$$= \frac{1}{24} \pi R^2 H$$

$$\frac{\text{Volume of upper part}}{\text{Volume of cone}} = \frac{\frac{1}{24} V}{V} = \frac{1}{24}$$

15. (b); Volume of spherical shell = $\frac{4}{3} \pi (R^3 - r^3)$

$$= \frac{4}{3} \times \pi (5^3 - 4^3) = \frac{4}{3} \times \pi \times 61 = \frac{244\pi}{3} \text{ cm}^3$$

16. (a); Surface area = $4\pi r^2 = 144\pi \text{ cm}^2$

$$r^2 = \frac{144\pi}{4\pi} = 36, \quad r = 6$$

$$\text{Volume} = \frac{4}{3}\pi \times 6 \times 6 \times 6 = 288\pi \text{ m}^3$$

17. (c); Volume of sphere = Volume of cylinder

$$\frac{4}{3}\pi R^3 = \pi \times 3 \times 3 \times 32$$

$$\Rightarrow R^3 = 216 \Rightarrow R = 6$$

18. (d); Volume of cylinder = Volume of sphere

$$\pi r^2 h = \frac{4}{3}\pi r^3$$

$$\frac{h}{r} = \frac{4}{3}$$

Ratio of their curved surface area

$$= \frac{2\pi r h}{4\pi r^2} = \frac{1}{2} \left(\frac{h}{r} \right) = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$$

19. (a); Ratio of whole surface of cone, hemisphere and cylinder

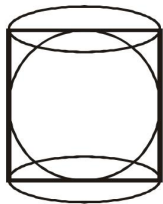
$$= \left(\pi R \left(\sqrt{R^2 + H^2} \right) + \pi R^2 \right) : 3\pi R^2 : 2\pi R H + 2\pi R^2$$

$$= \left(\pi H \sqrt{2H^2} + \pi H^2 \right) : 3\pi H^2 : 4\pi H^2$$

$$= \left(\sqrt{2} + 1 \right) \pi H^2 : 3\pi H^2 : 4\pi H^2$$

$$= \left(\sqrt{2} + 1 \right) : 3 : 4$$

20. (c);



Let radius of sphere = r

radius of cylinder = r

Height of cylinder = h

$$\frac{\text{Volume of sphere}}{\text{Volume of Cylinder}} = \frac{\frac{4}{3}\pi r^3}{\pi r^2 \times 2r} = \frac{2}{3} \text{ i.e. } 2:3$$

21. (b); Height of cone, $h = \frac{1}{2}r$

radius of sphere, $r = 5 \text{ cm}$

$$\text{Height, } h = \frac{1}{2} \times 5 = \frac{5}{2}$$

$$\frac{1}{3}\pi r_1^2 h = \frac{4}{3}\pi r^3$$

$$r_1^2 = \frac{4 \times 5 \times 5 \times 5 \times 2}{5} = 200$$

$$r_1 = \sqrt{200} = 10\sqrt{2} \text{ cm}$$

22. (c); Let the radius of the rod be R .

Then, its length = $12R$

Number of balls

$$\frac{\text{Volume of rod}}{\text{Volume of 1 ball}} = \frac{\pi R^2 \times 12R}{\frac{4}{3}\pi R^3} = 9$$

23. (b); Radius of cone = Radius of hemisphere = R .

Then, $2\pi R^2 = \pi R \sqrt{R^2 + H^2}$, where H is the height of cone

$$2R = \sqrt{R^2 + H^2} \Rightarrow 4R^2 = R^2 + H^2 \Rightarrow 3R^2 = H^2$$

$$\left(\frac{R}{H} \right)^2 = \frac{1}{3} \quad R:H = 1:\sqrt{3}$$

24. (b); Let R be the radius of the hemisphere and H be the height of the cylinder $\therefore R = H$

Ratio of the volume of cylinder, hemisphere and cone.

$$= \pi R^2 \times R : \frac{2}{3}\pi R^3 : \frac{1}{3}\pi R^2 \times R$$

$$= 1 : \frac{2}{3} : \frac{1}{3} = 3:2:1$$

25. (d); Let original volume = $l \times b \times h = V$

New volume = $2l \times 2b \times 2h = 8V$

So the volume becomes 8 times.

26. (a); Let the initial height be h and the height after sand is thrown be H metres.

Here we have to find $(H - h)$

According to question

$$12 \times 5 \times (H - h) = 210$$

$$(H - h) = \frac{210}{60} = 3.5 \text{ metres}$$

27. (b); $6a^2 = 24$, $a^2 = 4$, $a = 2 \text{ cm}$

$$\text{Number of cubes} = \left(\frac{100}{2} \right)^3 = (50)^3 = 125000$$

28. (a); $6a^2 = 96$, $a^2 = 16$, $a = 4$

Volume of small cubes = $a^3 = 4^3 = 64 \text{ cm}^3$

For larger cube

$$6A^2 = 384, \quad A^2 = 64, \quad A = 8$$

$$\text{Number of small cubes} = \frac{8 \times 8 \times 8}{64} = 8$$

29. (b); The radii of ends of the frustum are $\frac{48}{2\pi}$, $\frac{34}{2\pi}$

$$\text{Volume} = \frac{\pi}{3} \times 10 \times \left\{ \frac{48 \times 48}{(2\pi)^2} + \frac{34 \times 34}{(2\pi)^2} + \frac{48 \times 34}{(2\pi)^2} \right\}$$

$$= \frac{10 \times 7 \times 5092}{22 \times 12} = 1350$$

30. (a); Radius of coin = $r = \frac{1.5}{2} = \frac{3}{4}$ cm

$$\text{Thickness of the coin} = h = 2\text{mm} = \frac{1}{5} \text{ cm}$$

$$\therefore \text{Volume of one coin} = \pi r^2 h = \pi \left(\frac{3}{4}\right)^2 \left(\frac{1}{5}\right) = \frac{9\pi}{80}$$

$$\text{Radius of cylinder} = R = 3 \text{ cm}$$

$$\text{Height of cylinder} = H = 8 \text{ cm}$$

$$\text{Volume of cylinder} = \pi \times 3^2 \times 8 = 72\pi \text{ cm}^3$$

$$\text{Number of coins} = \frac{72\pi}{\frac{9\pi}{80}} = 640$$

31. (b); $2\pi rh + 2\pi r^2 = 231$

$$\text{And } 2\pi rh = \frac{2}{3} \times 231 = 154$$

$$2\pi r^2 = 77$$

$$r^2 = \frac{77}{2 \times 22} \times 7 = \frac{7 \times 7}{4}, \quad r = \frac{7}{2} \text{ cm}$$

$$h = \frac{154 \times 7 \times 2}{2 \times 22 \times 7} = 7 \text{ cm}$$

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 7 = \frac{539}{2}$$

$$= 269.5 \text{ cm}^3$$

32. (c); Volume of cylinder = $\pi \times 4^2 \times 3 = 48\pi$

$$\text{Volume of cone} = 48\pi$$

$$\frac{1}{3} \pi \times 4 \times 4 \times H = 48\pi$$

$$H = 9 \text{ cm}$$

33. (c); Curved surface area of tent = πrl

$$= \frac{22}{7} \times 6 \times 6.3 \text{ m}^2 = 118.8 \text{ m}^2$$

$$\text{Length of canvas} = \frac{118.8}{2} = 59.4 \text{ m}$$

34. (a); Radius of the cone = $r + 20\%$ of $r = 1.2r$ cm

$$\text{And slant height} = 2l \text{ cm}$$

$$\text{Surface area of new cone} = \pi \times 1.2r \times 2l = 2.4\pi rl$$

$$\text{Original Surface area} = \pi rl$$

$$\text{Increase in surface area} = 2.4\pi rl - \pi rl = 1.4\pi rl$$

$$\text{Percentage increase} = \frac{1.4\pi rl}{\pi rl} \times 100 = 140\%$$

35. (a); Volume of rectangular block = $10 \times 5 \times 2 = 100 \text{ cm}^3$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 7$$

$$= 66 \text{ cm}^3$$

$$\text{Percentage of wasted wood} = \frac{34}{100} \times 100 = 34\%$$

36. (a); Length of rectangular sheet = $\frac{264}{11} = 24 \text{ cm}$

When sheet is rolled along its breadth, the breadth becomes circumference of cylinder.

$$2\pi r = 11$$

$$r = \frac{11}{2 \times 22} \times 7 = \frac{7}{4} \text{ cm}$$

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times \frac{7}{4} \times \frac{7}{4} \times 24 = 231 \text{ cm}^3$$

37. (c); Radius of sphere = $\frac{d}{2} = \frac{6}{2} = 3 \text{ cm}$

$$\text{radius of wire } (r_1) = \frac{0.2}{2} = 0.1 \text{ cm}$$

$$\text{Volume of sphere} = \text{Volume of cylinder}$$

$$\frac{4}{3} \pi r^3 = \pi r_1^2 h$$

$$\frac{4}{3} \times 3 \times 3 \times 3 = .1 \times .1 \times h$$

$$h = \frac{36}{0.01} = 3600 \text{ cm} = 36 \text{ m}$$

38. (a); For percentage change in volume

$$= \left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2} \right] \%$$

$$\text{Here, } x = 0, \quad y = 0 \quad \text{and} \quad z = 8$$

$$= \left[0 + 0 + 8 + \frac{0}{100} + \frac{0}{(100)^2} \right] \% = 8\%$$

Volume of cylinder increases by 8%

39. (c); Volume of cone = $\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(50)^2 \times 25$

Volume of sphere = $\frac{4}{3}\pi r^3$

$\frac{1}{3}\pi \times 50 \times 50 \times 25 = \frac{4}{3}\pi r^3$

$r^3 = 25 \times 25 \times 25, r = 25$

Surface area of sphere = $4\pi r^2$

$= 4 \times 25 \times 25 \times \frac{22}{7}$

$= 7857.14 \text{ cm}^2$

40. (a); Volume of water collected in the tank in 8 hours
 $= 30 \times 20 \times 1 = 600 \text{ m}^3$

Volume of water collected in the tank in 1 hour

$= \frac{600}{8} = 75 \text{ m}^3$

Water comes through a pipe of cross-section

$= \frac{5 \times 5}{100 \times 100} = \frac{25}{10000} \text{ m}^2$

Speed of water = $\frac{75 \times 10000}{25} = 30000 \text{ m/h}$

$= 30 \text{ km/h}$

Difficult

1. (a); Volume of all these cubes = $4^3 + 5^3 + 6^3$
 $= 64 + 125 + 216 = 405 \text{ cm}^3$.

62 cm^3 is lost due to improper handling

Volume of new cube = $405 - 62 = 343 \text{ cm}^3$

Let the side of new cube be a cm.

$a^3 = 343, a = \sqrt[3]{343} = 7 \text{ cm}$

whole surface area = $6a^2 = 6 \times 7 \times 7 = 294 \text{ cm}^2$.

2. (c); Total surface area of hemisphere = $3\pi r^2 = 1848$

$r^2 = \frac{1848 \times 7}{3 \times 22} = 196, r = 14 \text{ cm}$

According to question

Volume of hemisphere = Volume of cone

$\frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h, 2r = h$

$h = 2 \times 14 = 28 \text{ cm}$

3. (a); Volume of cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \pi \times 6 \times 6 \times 24 = 288\pi$

Volume of sphere = $\frac{4}{3}\pi \times R^3$

$288\pi = \frac{4}{3}\pi \times R^3$

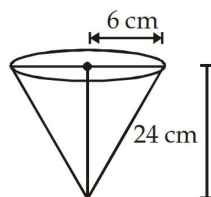
$R^3 = 216 \text{ cm}^3, R = 6 \text{ cm}$

4. (b); Diameter of beaker = 7 cm

then, Radius = $\frac{7}{2} \text{ cm}$

Diameter of marble = 1.4 cm

radius = $\frac{1.4}{2} = 0.7 \text{ cm}$



Let number of marbles dropped are 'n'.

Volume of n marble's dropped =

Volume of beaker up to 5.6 cm

$n \times \frac{4}{3}\pi \times 0.7 \times 0.7 \times 0.7 = \pi \times \frac{7}{2} \times \frac{7}{2} \times 5.6$

$n = \frac{7}{2} \times \frac{7}{2} \times 5.6 \times \frac{3}{4} \times \frac{10}{7} \times \frac{10}{7} \times \frac{10}{7} = 150$

5. (a); Edge of the cube $\sqrt[3]{343} = 7 \text{ cm}$

$\therefore$ radius of cone = 3.5 cm and height = 7 cm

$\therefore$ volume of the cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 7 = \frac{1}{3} \times 22 \times 12.25 \approx 90 \text{ cc.}$

6. (d); $c = \pi r l$
 $c^2 = \pi^2 r^2 l^2$

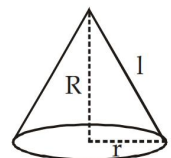
$V = \frac{1}{3}\pi r^2 h$

$V^2 = \frac{1}{9}\pi^2 r^4 h^2$

$3\pi v h^3 - c^2 h^2 + 9v^2 = 3\pi \times \frac{1}{3}\pi r^2 h \times h^3 - \pi^2 r^2 l^2 h^2$
 $+ 9 \times \frac{1}{9}\pi^2 r^4 h^2$

$= \pi^2 r^2 h^4 - \pi^2 r^2 h^2 (r^2 + h^2) + \pi^2 r^4 h^2$

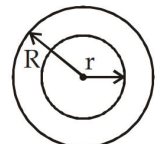
$= \pi^2 r^2 h^4 - \pi^2 r^4 h^2 - \pi^2 r^2 h^4 + \pi^2 r^4 h^2 = 0$



7. (d); Volume of lead = $\frac{4}{3}\pi r^3$

Volume of gold = $\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3$

$\frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3, \frac{4}{3}\pi R^3 = \frac{8}{3}\pi r^3$



$$R^3 = 2r^3, \quad R^3 = 2 \times 2^3$$

$$R = \sqrt[3]{2} \times 2 = 1.259 \times 2 = 2.518$$

$$\text{Thickness} = R - r = 2.518 - 2 = 0.518 \text{ cm.}$$

8. (c); $(l^2 + b^2 + h^2) = d^2$
 $\therefore (l + b + h)^2 = s^2$
 $\Rightarrow (l^2 + b^2 + h^2) + 2(lb + bh + lh) = s^2$
 $\Rightarrow 2(lb + bh + lh) = (s^2 - d^2) = \text{Surface area}$

9. (a); $\triangle ABC$ is equilateral triangle.
 $\angle ACB = 60^\circ$

$$\text{and } \angle BCP = 30^\circ$$

$$\triangle CDO, \angle CDO = 90^\circ$$

$$OD = 1 \text{ cm}$$

$$OC = 2 \text{ cm} \quad \left[\therefore \sin 30^\circ = \frac{OD}{OC} \right]$$

$$CZ = OC + OZ$$

$$= 2 + 1 = 3 \text{ cm}$$

$$\triangle CZY, \angle CZY = 90^\circ$$

$$CZ = 3 \text{ cm}$$

$$YZ = \sqrt{3}$$

$$\text{In cone } CZY$$

$$r = ZY = \sqrt{3} \text{ cm}$$

$$h = CZ = 3 \text{ cm}$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (\sqrt{3})^2 (3) = 3\pi \text{ cm}^3$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \text{ cm}^3$$

$$\text{Volume of water that can immerse the ball}$$

$$= \left(3\pi - \frac{4\pi}{3} \right) = \frac{5\pi}{3}$$

10. (a); Diameter = 5 mm = 0.5 cm
radius = 0.25 cm
Volume of water flowing from the pipe in one min.

$$= \pi \times 0.25 \times 0.25 \times 1000 = 62.5 \pi \text{ cm}^3$$

$$\text{Volume of conical vessel} = \frac{1}{3} \pi \times 15 \times 15 \times 24$$

$$\text{Time taken} = \frac{\frac{1}{3} \pi \times 15 \times 15 \times 24}{\pi \times 62.5}$$

$$= 28 \frac{4}{5} = 28 \text{ min } 48 \text{ sec.}$$

11. (a); Cone radius : height
4 : 3

$$\text{Let the radius and height be } 4x \text{ and } 3x.$$

$$\text{Curved surface area of cone} = \pi r l$$

$$l = \sqrt{h^2 + r^2}, \quad l = \sqrt{(3x)^2 + (4x)^2} = 5x$$

$$\text{Curved surface area} = \pi \times 4x \times 5x = 20\pi x^2$$

$$\text{Total surface area} = \pi r l + \pi r^2$$

$$= \pi r(l + r) = \pi \times 4x(5x + 4x)$$

$$= \pi \times 4x \times 9x = 36\pi x^2$$

$$\text{curved area : total area}$$

$$20\pi x^2 : 36\pi x^2$$

$$5 : 9$$

12. (d); Let r be radius of each. Let h and H be the heights of the cylinder and cone respectively. Then,

$$\pi r^2 h = \frac{1}{3} \pi r^2 H \Rightarrow \frac{h}{H} = \frac{1}{3}.$$

13. (d); Radius of larger sphere = R units

$$\text{Its volume} = \frac{4}{3} \pi R^3$$

$$\text{volume of one of the smaller cone made by larger}$$

$$\text{sphere} = \frac{1}{3} \pi R^3$$

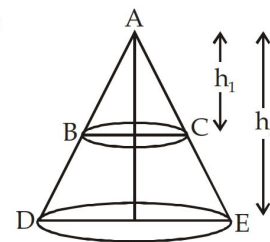
$$\text{Let the radius of smaller sphere is } r$$

$$\frac{4}{3} \pi r^3 = \frac{1}{3} \pi R^3, \quad \frac{r^3}{R^3} = \frac{1}{4}, \quad \frac{r}{R} = \frac{1}{\sqrt[3]{4}}$$

$$\frac{\text{surface area of smaller sphere}}{\text{Surface area of larger sphere}} = \frac{r^2}{R^2}$$

$$= \left(\frac{1}{(\sqrt[3]{4})^2} \right)^2 = \frac{1^2}{(2^{2/3})^2} = \frac{1}{2^{4/3}} = 1 : 2^{4/3}$$

14. (c);



$$\frac{\text{Volume of Cone ABC}}{\text{Volume of BCED}} = \frac{1}{1}$$

$$\frac{\text{Volume of Cone ABC}}{\text{Volume of Cone ADE}} = \frac{1}{1+1} = \frac{1}{2}$$

$$\left(\frac{\text{height of cone } (h_1)}{\text{height of cone } (h_2)} \right)^3 = \frac{1}{2}$$

$$\frac{h_1}{h_2} = \frac{1}{\sqrt[3]{2}} \quad h_1 : h_2 - h_1 = 1 : (\sqrt[3]{2} - 1)$$

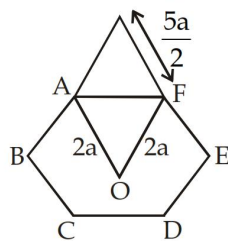
15. (a); Volume of coffee = $\frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 4^3$
 $= \frac{128}{3}\pi \text{ cm}^3$

Volume of cone = $\frac{1}{3}\pi r^2 \times h = \frac{1}{3} \times \pi \times 8^2 \times 16$
 $= \frac{1024}{3}\pi$

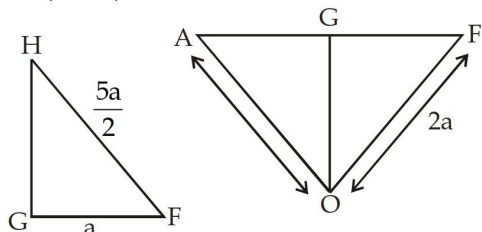
Required percentage = $\frac{\frac{1024}{3} - \frac{128}{3}}{\frac{1024}{3}} \times 100 = 87.5\%$

16. (c); Side of regular hexagon = $2a$

Area of hexagon = $6 \times \frac{\sqrt{3}}{4} \times (2a)^2 = 6\sqrt{3} a^2$



Slant edge of pyramid = $\frac{5a}{2}$
 (Given)



$HG = \sqrt{\left(\frac{5a}{2}\right)^2 - a^2} = \sqrt{\frac{25a^2}{4} - a^2} = \frac{\sqrt{21}}{2} a$

Altitude $GO = \frac{\sqrt{3}}{2} \times 2a = \sqrt{3} a$

Slant height = $\frac{\sqrt{21}}{2} a$, altitude = $\sqrt{3} a$

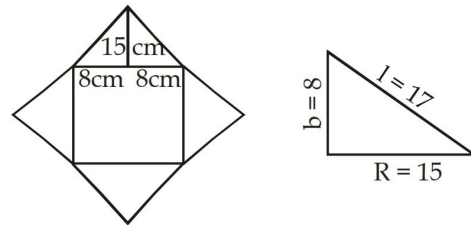
Height of pyramid = $\sqrt{\left(\frac{\sqrt{21}a}{2}\right)^2 - (\sqrt{3}a)^2}$

$= \sqrt{\frac{21a^2}{4} - 3a^2} = \frac{3}{2} a$

Volume of pyramid = $\frac{1}{3} \times \text{area of base} \times \text{height}$

$= \frac{1}{3} \times 6\sqrt{3} a^2 \times \frac{3}{2} a = 3\sqrt{3} a^3 \text{ cm}^3$

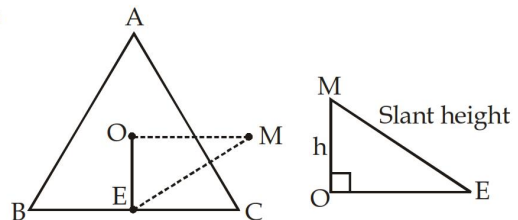
17. (b); Perimeter of base = $4 \times 16 = 64 \text{ cm}$
 Lateral surface area of pyramid = $\frac{1}{2} \times (\text{perimeter of base}) \times \text{slant height}$



Slant height = $\sqrt{15^2 + 8^2} = 17 \text{ cm}$

lateral surface area of pyramid = $\frac{1}{2} \times 64 \times 17$
 $= 544 \text{ cm}^2$

18. (d);



BC = Base of equilateral triangle.
 OE = Inradius of equilateral triangle.

$OE = \frac{\text{side of equilateral triangle}}{2\sqrt{3}}$

$= \frac{10\sqrt{3}}{2\sqrt{3}} = 5 \text{ cm}$

Slant length = $l = \sqrt{h^2 + OE^2} = \sqrt{h^2 + 25}$

Total surface area = $270\sqrt{3}$

Total surface area =

$\frac{1}{2} \times (30\sqrt{3} \times \sqrt{h^2 + 25}) + \frac{\sqrt{3}}{4} (10\sqrt{3})^2$

$270\sqrt{3} = \frac{1}{2} \{30\sqrt{3} \times \sqrt{h^2 + 25}\} + \frac{\sqrt{3}}{4} (10\sqrt{3})^2$

$\sqrt{h^2 + 25} = 13, \quad h^2 + 25 = 169$

$h^2 = 144, h = \sqrt{144} = 12 \text{ cm}$

19. (d); Let the side of square = $a \text{ cm}$

ATQ

T.S.A = C.S.A + 2 base area

= base perimeter $\times h + 2 \times \text{base area}$

$192 = 4a \times 10 + 2a^2$

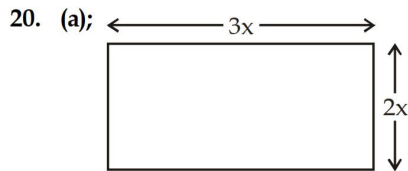
$a^2 + 20a - 96 = 0$

$$a(a + 24) - 4(a + 24) = 0$$

$$(a - 4)(a + 24) = 0 \quad a = 4, -24$$

$$\text{Volume} = \text{base area} \times h$$

$$\text{Volume} = 16 \times 10 = 160 \text{ cm}^3$$



Let length and breadth be $3x$ and $2x$.
 perimeter of base $= 2(3x + 2x) = 10x$
 area of base $= 2x \times 3x = 6x^2$
 height of prism $= 12 \text{ cm}$
 Total surface area $= \text{Perimeter of base} \times \text{height} + 2 \times \text{area of base}$
 $288 = 10x \times 12 + 12x^2$
 $12x^2 + 120x - 288 = 0$
 $x^2 + 10x - 24 = 0$
 $x = 2$
 area of base $= 6 \times 4 = 24 \text{ cm}^2$
 Volume of prism $= 24 \times 12 = 288 \text{ cm}^3$

21. (a); Total volume $= \pi r^2 h_1 + \frac{1}{3} \pi r^2 h_2$

A diagram of a cylinder with a cone on top. The cylinder's radius is labeled 21 cm and its height is 80 cm . The cone's height is 16 cm .

$$= \pi r^2 \left(h_1 + \frac{h_2}{3} \right)$$

$$= \frac{22}{7} \times (21)^2 \left(80 + \frac{16}{3} \right)$$

$$= \frac{22}{7} \times 441 \times \left(\frac{256}{3} \right)$$

$$= 22 \times 63 \times \frac{256}{3} = 22 \times 21 \times 256 = 118272 \text{ cm}^3$$

$$\text{Weight} = 118272 \times \frac{8.45}{1000} = 999.39 \text{ kg}$$

22. (a); Total surface area $= 2\pi R^2 + 2\pi r^2 + (\pi R^2 - \pi r^2)$
 $= 3\pi R^2 + \pi r^2 = \pi(3R^2 + r^2)$

$$1436 \frac{2}{7} = \pi(3R^2 + r^2)$$

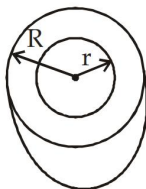
$$\frac{10054}{7} \times \frac{7}{22} = 3(12)^2 + r^2$$

$$457 = 432 + r^2$$

$$r^2 = 457 - 432$$

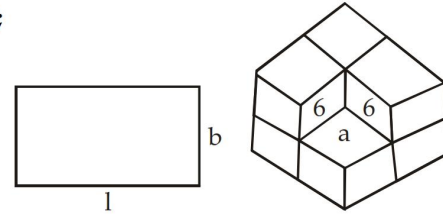
$$r^2 = 25, r = 5 \text{ cm}$$

$$\text{Internal volume of Hemisphere} = \frac{2}{3} \pi (R^3 - r^3)$$



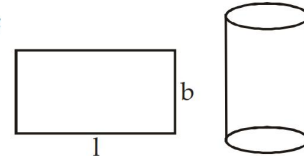
$$= \frac{2}{3} \times \frac{22}{7} (12^3 - 5^3) = 3358 \frac{2}{3} \text{ cm}^3$$

23. (d);



Since, three are 3 faces which are visible in the corner cube. When the cube of a corner is removed then the 3 faces of other cube will be visible from outside. So, there will be no change in the surface area of this solid figure.

24. (a);



$$\text{Length, } l = 2\pi r \quad r = \frac{l}{2\pi}$$

$$\text{Volume of cylinder} = \pi r^2 h = \pi \times \left(\frac{l}{2\pi} \right)^2 \times b$$

$$\frac{\pi \times l^2 b}{4\pi^2} = 48.125 = \frac{385}{8}$$

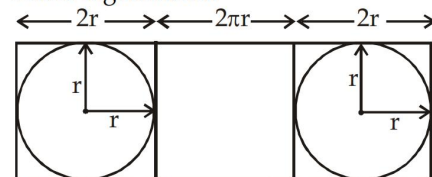
$$l^2 b = 11 \times 11 \times 5$$

$$l = 11 \text{ and } b = 5$$

$$\text{Volume of box} = (l - 1) \times (b - 1) \times 0.5$$

$$= 10 \times 4 \times 0.5 = 20 \text{ cm}^3$$

25. (a); For minimum wastage he has to cut in the following manner..



$$\text{Total area of sheet required} = (2\pi r + 4r) \times 2r$$

$$= 4r^2 (\pi + 2)$$

$$\text{Area of sheet utilized} = 2\pi r \times 2r + 2(\pi r^2) = 6\pi r^2$$

$$\text{Area of wastage sheet} = 4r^2 (\pi + 2) - 6\pi r^2$$

$$= 8r^2 - 2\pi r^2$$

$$\text{Required ratio} = \frac{8r^2 - 2\pi r^2}{6\pi r^2} = \frac{2r^2 (4 - \pi)}{6r^2 \pi} = \frac{1}{11}$$

26. (c); The required capacity of box is 864 m^3 . Let the length of box be 'l' and height of the box be 'h'.

$$\text{then, } 864 = l^3 h \Rightarrow h = \frac{864}{l^3}$$

$$\text{Now, surface area of box } A = l^2 + 4lh$$

$$= l^2 + \frac{4l \times 864}{2} = l^2 + \frac{3456}{l}$$

Differentiating with respect to l we get

$$\frac{dA}{dl} = 2l - \frac{3456}{l^2}$$

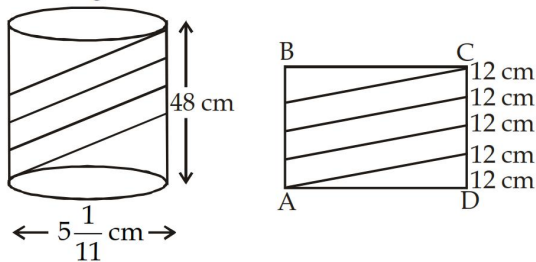
$$\text{For minimum area} = \frac{dA}{dl} = 0$$

$$2l = \frac{3456}{l^2} \Rightarrow l^3 = 1728, l = 12$$

$$\text{Base area} = 12^2 = 144 \text{ m}^2$$

$$\text{Height} = \frac{864}{l^2} = \frac{864}{144} = 6 \text{ m}$$

27. (d); According to the given information, the string will be bowded on a cylindrical tank as shown in figure.

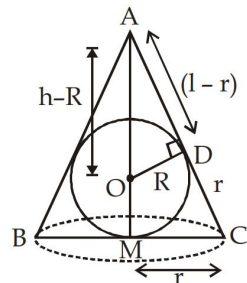


$$\text{The base circumference} = 2\pi r = 2 \times \frac{22}{7} \times \frac{56}{11} \times \frac{1}{2} = 16 \text{ cm}$$

$$AM = \text{length of one complete turn} = \sqrt{16^2 + 12^2} = 20 \text{ cm}$$

$$\text{Total length} = 4 \times 20 = 80 \text{ cm}$$

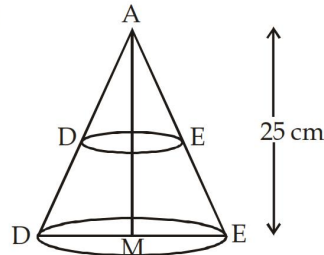
28. (c);



Let the radius of cone = r
and height = h , slant height = l
and radius of sphere = R

$$R = \frac{hr}{l+r} = \frac{8 \times 6}{10+6} = 3 \text{ cm}$$

29. (a);



$$\triangle AME \sim \triangle ANC$$

$$\frac{25-h}{25} = \frac{r}{5}$$

$$h = 25 - 5r \quad \dots (i)$$

$$\text{Value of frustum (v)} = \frac{1}{3}\pi[5^2 + r^2 + 5r]h$$

$$110 = \frac{1}{3}\pi[25 + r^2 + 5r](25 - 5r)$$

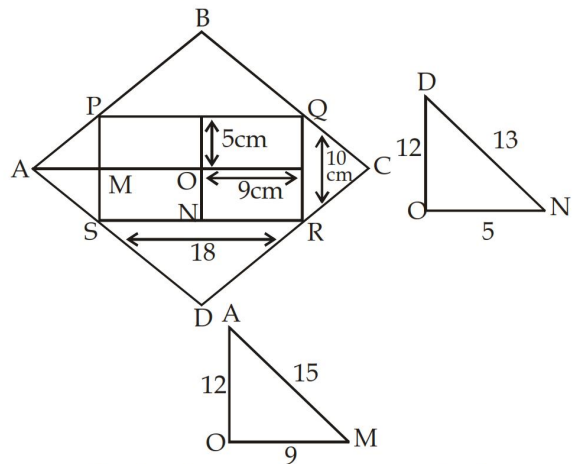
$$\frac{5}{3}\pi[(5-r)(5^2 + r^2 + 5r)] = 110$$

$$\frac{5}{3}\pi[5^3 - r^3] = 110, \quad 5^3 - r^3 = \frac{110 \times 3}{5\pi}$$

$$125 - r^3 = \frac{110 \times 3}{5 \times \frac{22}{7}}, \quad r^3 = 104, \quad r = (104)^{1/3}$$

30. (b); In $\triangle AOM$

$$\text{Slant height, } AM = \sqrt{12^2 + 9^2} = 15 \text{ cm}$$



$$\text{Slant height of } \triangle PAS = \text{slant height of } \triangle QCR$$

$$\text{Similarly slant height } \triangle PBQ = \text{slant height of } \triangle SDR$$

$$\text{Slant height } DN = \sqrt{12^2 + 5^2} = 13 \text{ cm}$$

$$\text{Slant height of } DM = \sqrt{12^2 + 9^2} = 15 \text{ cm}$$

$$\text{Area of } \triangle APS = \frac{1}{2} \times 10 \times 15 = 75 \text{ cm}^2$$

$$\text{Area of } \triangle DRS = \frac{1}{2} \times 18 \times 13 = 117 \text{ cm}^2$$

$$\text{Total surface area} = \text{Base area} + \text{area of } (\triangle APS + \triangle PBQ + \triangle CQR + \triangle DRS)$$

$$= 180 + 2 \times 117 + 2 \times 75 = 564 \text{ cm}^2$$

Previous Year (Memory Based)

1. (c); Each side of cube = a unit
Volume of cube = a^3

Now, new side of cube = $a \times \frac{110}{100} = 1.1a$ unit
New volume = $(1.1a)^3 = 1.331a^3$
Required Increase percentage
$$= \frac{1.331a^3 - a^3}{a^3} \times 100 = 33.1\%$$
2. (d); Surface area of small cube = $6 \times (\text{edge})^2$
 $= 6a^2 = 6 \text{ cm}^2$
Surface area of large cube = $6 \times (5)^2 = 150 \text{ cm}^2$

Required ratio = $\frac{6}{150} = 1 : 25$
3. (d); Let the sides are x, 2x and 4x respectively and the edge of cube be 'a'.
According to question
 $x \times 2x \times 4x = a^3, \quad 8x^3 = a^3$
Taking cube root, $a = 2x$
Surface area of parallelepiped = $2(lb + bh + hl)$
 $= 2(x \times 2x + 2x \times 4x + 4x \times x)$
 $= 2(2x^2 + 8x^2 + 4x^2) = 28x^2$ units
Surface area of cube = $6a^2 = 6(2x)^2 = 24x^2$

Required ratio = $\frac{28x^2}{24x^2} = \frac{28}{24} = 7 : 6$
4. (c); Area of four walls = $2(l + b) \times h$
 $= 2(15 + 12) \times h = 54h$
Now, area of ceiling + floor = $2 \times 15 \times 12 = 360$

 $54h = 360, \quad h = \frac{360}{54} = \frac{20}{3} \text{ m}$
Volume of hall = $l \times b \times h$
$$= 15 \times 12 \times \frac{20}{3} = 1200 \text{ m}^3$$
5. (d); Let the length, breadth and height are x, 2x and 3x.
$$\begin{array}{l} x \xrightarrow{100\% \uparrow} 2x \\ 2x \xrightarrow{200\% \uparrow} 6x \\ 3x \xrightarrow{300\% \uparrow} 12x \end{array}$$

New volume = $2x \times 6x \times 12x = 144x^3$
Increase in volume = $144x^3 - 6x^3 = 138x^3$

Increase in volume = $\frac{138x^3}{6x^3} = 23$ times
6. (c); Capacity of cistern = $(l - 2t)(b - 2t)(h - x)$
here $l = 330 \text{ cm}, b = 260 \text{ cm}, h = 110 \text{ cm}, t = 5 \text{ cm}$
$$8000 = \frac{(330 - 10)(260 - 10)(110 - x)}{1000}$$

$$110 - x = \frac{8000 \times 1000}{320 \times 250} = 100$$

$$x = (110 - 100) \text{ cm} = 10 \text{ cm} = 1 \text{ dm}$$
7. (d); Number of boxes = $\frac{800 \times 700 \times 600}{8 \times 7 \times 6} = 1000000$
8. (a); Volume of solid sphere = $\frac{4}{3}\pi r^3$
$$= \frac{4}{3} \times \pi \times 3 \times 3 \times 3 = 36\pi \text{ cm}^3$$

Let the radius of small sphere be 'x'.
$$36\pi = 8 \times \frac{4}{3}\pi x^3$$

$$x^3 = \frac{36}{8} \times \frac{3}{4}, \quad x^3 = \frac{27}{8}, \quad x = \frac{3}{2} = 1.5 \text{ cm}$$
9. (a); $\frac{\text{Volume of sphere 1}}{\text{Volume of sphere 2}} = \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{8}{27}$
$$\left(\frac{r_1}{r_2}\right)^3 = \left(\frac{2}{3}\right)^3, \quad \frac{r_1}{r_2} = \frac{2}{3}$$

$$\frac{\text{Surface area of sphere 1}}{\text{Surface area of sphere 2}} = \frac{4\pi r_1^2}{4\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$$
10. (d); Total surface area = $\pi rl + \pi r^2$
radius = 3cm, height = 4cm
slant height, $l = \sqrt{3^2 + 4^2} = 5 \text{ cm}$
$$\text{T.S.A} = \frac{22}{7} \times 3 \times 5 + \frac{22}{7} \times 3 \times 3$$

$$= \frac{22}{7} \times 3[5 + 3] = \frac{22}{7} \times 3 \times 8 \text{ cm}^2 = 75.4 \text{ cm}^2$$
11. (b); Volume of cone = 1232
$$\frac{1}{3}\pi r^2 h = 1232, \quad \frac{1}{3} \times \frac{22}{7} \times r^2 \times 24 = 1232$$

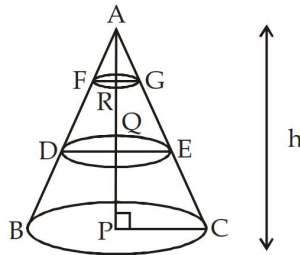
$$r^2 = \frac{1232 \times 3 \times 7}{22 \times 24} = 49, \quad r = \sqrt{49} = 7 \text{ cm}$$

$$\begin{aligned}\text{Slant height } l &= \sqrt{h^2 + r^2} = \sqrt{24^2 + 7^2} \\ &= \sqrt{576 + 49} = \sqrt{625} = 25 \text{ cm}\end{aligned}$$

Curved surface area

$$= \pi r l = \frac{22}{7} \times 7 \times 25 = 550 \text{ cm}^2$$

12. (d);



Let the radius of base of trisected part be

$$FR = r_1, DQ = r_2, BP = r_3$$

$$\text{and } AR = RQ = QP = \frac{h}{3}$$

Now, In $\triangle AFR$ and $\triangle ADQ$.

Both are similar triangle, then by using basic Proportionality Theorem

$$\frac{FR}{DQ} = \frac{AR}{AQ}, \frac{r_1}{r_2} = \frac{\frac{h}{3}}{\frac{2h}{3}} \Rightarrow \frac{r_1}{r_2} = \frac{1}{2}, r_2 = 2r_1$$

Similarly, In $\triangle AFR$ and $\triangle ABP$, $r_3 = 3r_1$

$$V_1 : V_2 : V_3$$

$$= \frac{1}{3} \pi r_1^2 \frac{h}{3} : \frac{1}{3} \pi (r_1^2 + r_2^2 + r_1 r_2) \frac{h}{3} : \frac{1}{3} \pi (r_1^2 + r_3^2 + r_1 r_3) \frac{h}{3}$$

$$= r_1^2 : (r_1^2 + 4r_1^2 + 2r_1^2) : (4r_1^2 + 9r_1^2 + 6r_1^2)$$

$$= r_1^2 : 7r_1^2 : 19r_1^2 = 1 : 7 : 19$$

13. (b); Let the radius of the base of cone be $5x$ cm and its height be $12x$ cm.

$$\text{Volume of cone} = 314 \frac{2}{7}$$

$$\Rightarrow \frac{1}{3} \pi r^2 h = \frac{2200}{7} \Rightarrow (5x)^2 (12x) = \frac{2200 \times 3 \times 7}{7 \times 22}$$

$$x^3 = \frac{100 \times 3}{25 \times 12}, x^3 = 1, x = 1 \text{ cm}$$

$$\text{Height} = 12x = 12 \times 1 = 12 \text{ cm}$$

$$\text{radius} = 5x = 5 \times 1 = 5 \text{ cm}$$

$$\text{slant height } l = \sqrt{h^2 + r^2} = \sqrt{12^2 + 5^2} = 13 \text{ cm}$$

14. (c); Radius of circular base = $\frac{19.2}{2} = 9.6 \text{ m}$

$$h = 2.8 \text{ m}$$

$$\begin{aligned}\text{Slant height} &= \sqrt{h^2 + r^2} = \sqrt{2.8^2 + 9.6^2} \\ &= \sqrt{7.84 + 92.16}, l = \sqrt{100} = 10 \text{ m}\end{aligned}$$

$$\text{Area of canvas} = \pi r l = \frac{22}{7} \times 9.6 \times 10 = 301.71 \text{ m}^2$$

15. (d); Volume of cone = $\frac{1}{3} \pi r^2 h$

Volume increase or decrease

$$= \left(x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{100 \times 100} \right)$$

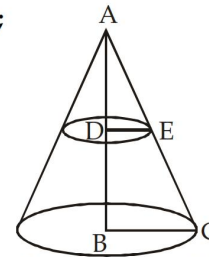
$$\text{Here } x = 200, y = -50, z = -50$$

$$= \left(200 - 50 - 50 + \frac{-10000 + 2500 - 10000}{100} + \frac{500000}{100 \times 100} \right)$$

$$= 100 + (-175) + 50 = -25$$

$$\text{Decrease in area} = 25\%$$

16. (d);



Since $\triangle ADE \sim \triangle ABC$

$$\frac{AD}{AB} = \frac{DE}{BC} = \frac{1}{2}$$

$$\therefore AD = \frac{1}{2} AB \text{ and } DE = \frac{1}{2} BC$$

$$\therefore \text{Required Ratio} = \frac{\text{Volume of cone}}{\text{Volume of frustum}}$$

$$= \frac{\frac{1}{3} \times \pi \times DE^2 \times AD}{\frac{1}{3} \times \pi \times BC^2 \times AB - \frac{1}{3} \pi DE^2 \times AD}$$

$$= \frac{DE^2 \times AD}{BC^2 \times AB - DE^2 \times AD}$$

$$= \frac{\frac{1}{4} BC^2 \times \frac{1}{2} AB}{BC^2 \times AB - \frac{1}{4} BC^2 \times \frac{1}{2} AB} = \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{1}{7} = 1:7$$

17. (a); Radius of base of A = $\frac{1}{2}r$, Height of A = $\frac{1}{2}h$.

$$\text{Volume of A} = \frac{1}{3}\pi \times \left(\frac{1}{2}r\right)^2 \times \left(\frac{1}{2}h\right) = \frac{1}{24}\pi r^2 h$$

$$\text{Volume of B} = \left(\frac{1}{3}\pi r^2 h - \frac{1}{24}\pi r^2 h\right) = \frac{7}{24}\pi r^2 h.$$

$\therefore \text{Vol A} < \text{Vol B}$

18. (c); Circumference = length of rectangular piece

$$2\pi r = 22, \quad 2 \times \frac{22}{7} \times r = 22, \quad r = \frac{7}{2} = 3.5 \text{ cm}$$

Now height of cylinder,

$h = \text{breadth of rectangular piece} = 12$

$\Rightarrow \text{Volume of cylinder} = \pi r^2 h$

$$= \frac{22}{7} \times 3.5 \times 3.5 \times 12 = 462 \text{ cm}^3$$

19. (a); Ratio of diameter = $\frac{d_1}{d_2} = \frac{r_1}{r_2} = \frac{3}{2}$

Since volume are equal

$$\pi r_1^2 h_1 = \pi r_2^2 h_2$$

$$\left(\frac{r_1}{r_2}\right)^2 = \frac{h_2}{h_1}, \quad \left(\frac{3}{2}\right)^2 = \frac{h_2}{h_1}, \quad h_1 : h_2 = 4 : 9$$

20. (b); Given, $\frac{r_1}{r_2} = \frac{3}{4}$ and $\frac{h_1}{h_2} = \frac{8}{7}$

$$\text{Required ratio} = \frac{\text{Volume of first cylinder}}{\text{Volume of second cylinder}}$$

$$= \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2}\right)^2 \times \left(\frac{h_1}{h_2}\right) = \left(\frac{3}{4}\right)^2 \times \frac{8}{7} = \frac{9}{14}$$

21. (c); Volume of iron used = $\pi h(R^2 - r^2)$

$$D = 8, \quad d = 6$$

$$R = 4, \quad r = 3$$

$$\text{Volume} = \frac{22}{7} \times 20 \times [4^2 - 3^2]$$

$$= \frac{22}{7} \times 20 \times 7 = 440 \text{ cm}^3$$

22. (a); Given, volume of pyramid = 500 m^3

$$\frac{1}{3} \times \text{Area of base} \times \text{Height} = 500$$

$$\frac{1}{3} \times 30 \times \text{Height} = 500$$

$$\therefore \text{Height} = \frac{500}{10} = 50 \text{ m}$$

23. (b); Area of base of prism = $\frac{1}{2} \times 10 \times 12 = 60 \text{ cm}^2$

Volume of prism = Area of base $\times$ Height

$$= 60 \times 20 = 1200 \text{ cm}^3$$

Weight of prism = Volume $\times$ Density

$$= \left(\frac{1200 \times 6}{1000}\right) \text{ kg} = 7.2 \text{ kg}$$

24. (b); Slant height of triangle

$$= \sqrt{15^2 + 8^2} = \sqrt{289} = 17 \text{ cm}$$

Lateral surface area

$$= \frac{1}{2} \times (\text{Perimeter of base}) \times \text{Slant height}$$

$$= \frac{1}{2} \times 17 \times 64 = 544 \text{ cm}^2$$

25. (c); Semi-perimeter of triangle = $S = \frac{9+12+15}{2}$

$$= 18 \text{ cm}$$

$$\text{Area of triangle} = \sqrt{S(S-a)(S-b)(S-c)}$$

$$= \sqrt{18 \times 9 \times 6 \times 3}$$

$$= \sqrt{9 \times 2 \times 9 \times 2 \times 3 \times 3} = 54 \text{ cm}^2$$

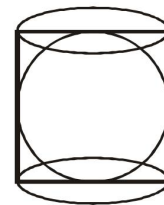
Total surface area = (Base Perimeter $\times$ height) + (2 $\times$ Area of base)

$$= (12 + 15 + 9) \times 5 + 2 \times 54$$

$$= 180 + 108 = 288 \text{ cm}^2$$

26. (a); Since, the sphere is inscribed in the cylinder and the radius of sphere = R

Radius of cylinder = Radius of sphere



Height of cylinder = 2 $\times$ Radius of sphere = 2R

Volume of cylinder = $\pi r^2 h$

$$= \pi R^2 \times 2R = 2\pi R^3$$

27. (a); Lateral surface area = Perimeter of the base $\times$ Height

$$120 = \text{Perimeter} \times \text{Height}$$

$$120 = 3 \times a \times h$$

...(i)

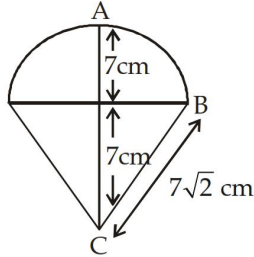
Volume = Area of base $\times$ height

$$40\sqrt{3} = \frac{\sqrt{3}}{4} a^2 \times h \quad \dots(ii)$$

On dividing equation (ii) by equation (i), we get

$$\frac{40\sqrt{3}}{120} = \frac{\frac{\sqrt{3}}{4} a^2 \times h}{3a \times h}, \quad \frac{\sqrt{3}}{3} = \frac{a}{\sqrt{3} \times 4}, \quad a = 4 \text{ cm}$$

28. (a);

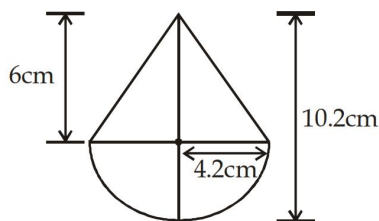


$$\begin{aligned} \text{Volume of ice-cream cone} &= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3 \\ &= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 + \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 \\ &= \frac{22}{3} (49 + 98) = \frac{22}{3} \times 147 = 22 \times 49 = 1078 \text{ cu.cm} \end{aligned}$$

29. (d); Let radius of the sphere = r cm.
Then, according to the question,
Volume of right circular cones = Volume of sphere

$$\begin{aligned} \frac{1}{3} \pi (2.1)^2 \times 3.8 + \frac{1}{3} \pi (2.1)^2 \times 4.6 &= \frac{4}{3} \pi r^3 \\ \frac{\pi}{3} (2.1)^2 (3.8 + 4.6) &= \frac{4}{3} \pi r^3, \quad 2.1^2 \times 8.4 = 4r^3 \\ r^3 &= 2.1^3, \quad r = 2.1 \text{ cm} \\ \text{diameter} &= 2r = 4.2 \text{ cm} \end{aligned}$$

30. (d);



$$\begin{aligned} \text{Volume of wooden toy} &= \frac{2}{3} \pi r^3 + \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi [2 \times (4.2)^3 + (4.2)^2 \times 6] \\ &= \frac{1}{3} \times \frac{22}{7} \times 4.2 \times 4.2 (8.4 + 6) \\ &= 22 \times 0.2 \times 4.2 \times 14.4 = 266 \text{ cm}^3 \end{aligned}$$

$$31. (b); \quad \frac{r_1}{r_2} = \frac{\sqrt{3}}{\sqrt{2}}, \quad \frac{h_1}{h_2} = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\begin{aligned} \text{Required ratio} &= \frac{\pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2} = 3 \times \left(\frac{r_1}{r_2} \right)^2 \times \left(\frac{h_1}{h_2} \right) \\ &= \frac{3}{1} \times \frac{3}{2} \times \frac{\sqrt{2}}{\sqrt{3}} = 3\sqrt{3} : \sqrt{2} \end{aligned}$$

32. (c); Let the initial height of water in Can is h
Final Volume of Can - volume of sphere = Initial Volume

$$\pi (3.5)^2 \times (2 \times 3.5) - \frac{4}{3} \pi (3.5)^3 = \pi (3.5)^2 h$$

$$2 \times 3.5 - \frac{4}{3} \times 3.5 = h, \quad 7 - \frac{14}{3} = h, \quad h = \frac{7}{3}$$

33. (d); Then, volume of the cone = $\frac{1}{3} \pi r^2 h$

$$\text{Total volume} = \pi r^2 h$$

$$\therefore \text{Volume of cylinder} = \left(\pi r^2 h - \frac{1}{3} \pi r^2 h \right) = \frac{2}{3} \pi r^2 h$$

Let H be the height of the cylinder.

$$\text{Then, } \pi r^2 H = \frac{2}{3} \pi r^2 h \Rightarrow H = \frac{2}{3} h.$$

34. (c); According to question
Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 10 \times 10 \times 21 = 6600 \text{ cm}^3$$

Let the height of the removed cone be (h)

$$6600 - \frac{1}{3} \pi \times 10^2 \times h = 4400$$

$$\frac{1}{3} \times \frac{22}{7} \times 10^2 \times h = 2200$$

$$h = \frac{2200 \times 3 \times 7}{22 \times 100} = 21 \text{ cm}$$

35. (d); Here, h = 8r

$$\text{Volume of cylinder} = \pi r^2 h = \pi r^2 (8r) = 8\pi r^3$$

$$\text{radius of sphere} = \frac{1}{2} r$$

Number of spherical balls

$$= \frac{8\pi r^3}{\frac{4}{3} \pi \frac{r^3}{8}} = \frac{8 \times 8 \times 3}{4} = 48$$

36. (b); Let the radius of sphere be 'x'

Side of cube be 'a'

According to question

$$4\pi r^2 = 6a^2, \quad \frac{r^2}{a^2} = \frac{6}{4\pi}, \quad \frac{r}{a} = \frac{\sqrt{3}}{\sqrt{2\pi}}$$

$$\frac{\text{Volume of sphere}}{\text{Volume of cube}} = \frac{\frac{4}{3}\pi r^3}{a^3} = \frac{4}{3}\pi \left(\frac{r}{a}\right)^3$$

$$= \frac{4}{3}\pi \frac{3\sqrt{3}}{2\pi\sqrt{2\pi}} = \frac{\sqrt{6}}{\sqrt{\pi}} = \sqrt{6} : \sqrt{\pi}$$

37. (b); Volume of copper wire = $\pi r^2 h$

$$= \pi \left(\frac{1}{10}\right)^2 \times 3600 = 36\pi$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\frac{4}{3}\pi r^3 = 36\pi, \quad r^3 = \frac{36 \times 3}{4} = 9 \times 3 = 27$$

$$r^3 = 27, \quad r = 3$$

38. (a); Let the edge of the cube be 'a'

$$\text{radius of sphere} = \frac{a}{2}$$

$$\frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{a^3}{\frac{4}{3}\pi \left(\frac{a}{2}\right)^3} = \frac{8a^3}{\frac{4}{3}\pi a^3} = 6 : \pi$$

39. (a); Height = 15 cm, diameter = 30 cm

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi (15)^3$$

$$\text{Volume of sphere} = \frac{4}{3}\pi (15)^3$$

Percentage of wasted wood

$$= \frac{\frac{1}{3}\pi (15)^3 (4-1)}{\frac{4}{3}\pi 15^3} \times 100 = \frac{3}{4} \times 100 = 75\%$$

40. (a); Volume of sphere = $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times 8^3 = \frac{512 \times 4}{3}\pi$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi 8^2 h$$

$$\frac{1}{3}\pi 8^2 h = \frac{512 \times 4}{3}\pi, \quad h = \frac{512 \times 4 \times 3}{3 \times 64} = 32 \text{ cm}$$

$$\text{Slant height} = \sqrt{h^2 + r^2}$$

$$= \sqrt{32^2 + 8^2} = \sqrt{1024 + 64} = \sqrt{1088} = 8\sqrt{17} \text{ cm}$$